International Islamic University Chittagong



Final Lab Report

Nutri-Track Academia: A Student's Daily-Meal Management System

Course Title: Software Engineering Lab

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Chapter 1

Introduction

1.1 Abstract:

Nutri-Track Academia is a Students' Daily Meal Management System that provides optimization of meal reservation, invoicing, and nutritional tracking procedures in the academic environment. The system assigns a unique meal card number to each student upon registration, allowing seamless transactions in dining areas. It further guarantees a personalized and balanced nutrition experience by utilizing the Body Mass Index (BMI) and Basal Metabolic Rate (BMR) calculators, which set personal daily caloric requirements.

With a structured meal planning process, Nutri-Track Academia collaborates with competent caterers to provide menu variations that are sensitive to the time of the day and individual health requirements, as well as to students' food preferences. This manages the entire ordering process along with the calculations of food bills. The caterers are capable of tracking daily orders effectively, thus providing an improved supply chain process. In addition, the system generates individual meal reports that allow students to evaluate their eating habits and make knowledgeable decisions about their food consumption. Prioritizing nutrition, convenience, and transparency, Nutri-Track Academia aims to provide an overall meal management solution for students.

1.2 Introduction:

Nutri-Track Academia is a planned website development project whose main focus is designing a "Students' Daily Meal Management System", preferably in a college/university setting. This project aims to create a fully functional website that will handle a student's daily meal planning and ordering activities. Functionality plans include generating a meal card for every individual student so that they can order their meals throughout their campus stay time using this card, options of calculating their BMI and BMR to suggest proper diet, proper meal menu display on the basis of time of the day, management of meal ordering and bill calculation, and so on. This website basically plans to be like a meal management assistant for a student, which will not only help them in purchasing meals at campus food outlets, but will also take care of their health by suggesting proper meal plans to each individual and displaying menus suited to each time of the day. The success of this project is believed to play an important role in ensuring the good health of students in their prime through proper meal management during a major part of their days.

1.3 Objectives:

1. Create a **fully functional meal planning and ordering platform** to streamline students' dining experiences.
2. Implement a **digital meal card system** that allows students to order meals conveniently throughout their campus stay.

3. Integrate **BMI and BMR calculators** to provide personalized dietary recommendations based on students' health metrics.
4. Display **dynamic meal menus** based on the time of day to help students make informed food choices, like showing breakfast and lunch menus at their specific times.
5. Facilitate **meal ordering and bill management** for a seamless transaction process.
6. Contribute to **efficient food service operations** within campus dining facilities.
7. Provide an **admin dashboard** to manage the entire meal system efficiently.
8. Allow **menu customization**, enabling administrators to update and modify meal options as needed.
9. Maintain **comprehensive order records** for all students, ensuring transparency and tracking meal usage.
10. Enable **order history tracking** to analyze meal consumption trends and improve inventory management.

1.4 Motivation:

Many students struggle with maintaining a healthy diet due to their demanding academic schedules, financial constraints, and lack of access to nutritious food. Balancing studies, extracurricular activities, and personal commitments often leads to irregular eating habits, reliance on unhealthy fast food, or even skipping meals altogether. These poor dietary choices can result in various health issues such as obesity, malnutrition, fatigue, and a decline in cognitive performance, ultimately affecting academic success and overall well-being.

The challenge is further intensified by the limited availability of affordable and nutritious meal options on campus. Students may find it difficult to plan meals that meet their nutritional needs while staying within budget. Additionally, a lack of awareness regarding proper nutrition and meal planning contributes to unhealthy eating patterns that persist beyond college life.

Furthermore, thinking from the employees of campus canteens' side, personally taking orders from each student, calculating the bills of each order, and keeping the meal order records of the students throughout a timeline can prove to be very hectic for employees of the canteens to do manually, especially if the number of students on campus is huge. There can also be errors in money management through the wrong calculation of bills. But computerizing the whole system can benefit the campus greatly by reducing the time and effort of employees and managing everything properly.

Recognizing these challenges, there is a growing need for structured solutions that provide students with accessible, affordable, nutritious meal options and manage the whole meal ordering and billing system. Addressing these issues can significantly enhance student health, energy levels, and overall academic performance, fostering a more productive and focused learning environment. And it can also reduce the work burden on the meal management personnel by creating such a systematic method. **Nutri-Track Academia** has been planned keeping all these factors in mind.

1.5 Literature Review:

➢ "A Smart Dietitian using BMR: Focused on Bangladeshi Food Habits"

- ✓ **BMI & BMR Calculation**: Uses BMI and BMR to personalize diet recommendations.
- ✓ **Diet Chart Generation**: Provides a customized meal plan based on calorie requirements.
- ✓ **Health Condition Consideration**: Takes into account diabetes, allergies, and other conditions.
- ✓ **Bangladeshi Food Habit Focus**: Tailors diet charts based on local food preferences.
- ✓ **Exercise & Activity Tracking**: Suggests exercises along with dietary recommendations.
- ✓ **User-Friendly Interface**: Allows easy input of personal details for diet customization.
- ✓ **Print & Download Options**: Generates a printable/downloadable diet plan for future reference.
- ✓ **Awareness & Guidance**: Educates users on healthy eating habits and lifestyle improvements.

➢ "Automation in Food Ordering System and Bill Payment in Hotels using TFT Display"

- ✓ **Touchscreen-Based Ordering**: Uses TFT displays at restaurant tables for menu browsing and order placement.
- ✓ **Wireless Technology Integration**: Incorporates Wi-Fi, RFID, GSM, and GPRS for seamless communication.
- ✓ **Automated Order Processing**: Orders are sent directly to the kitchen display system for preparation.
- ✓ **Dynamic Menu Management**: Allows real-time updates of menu items and prices.
- ✓ **Cashless Payment System**: Supports RFID-based prepaid payments and mobile transactions.
- ✓ **Customer Feedback Collection**: Provides a feedback system to assess service quality.
- ✓ **Faster Table Turnover**: Reduces waiting times for order placement and service.
- ✓ **Efficient Billing Process**: Automatically calculates and displays the bill at the end of the meal.
- ✓ **Kitchen Display System**: Chefs receive real-time order updates, improving efficiency in food preparation.
- ✓ **Restaurant Manager Dashboard**: Allows managers to monitor orders, update menus, and analyze sales.

➢ "Students' Meal Information System (SMIS)"

- ✓ **Digital Meal Card Integration**: Assigns each student a unique digital meal card, enabling seamless meal ordering and reducing wait times.
- ✓ **Efficient Meal Tracking**: Accurately records meal consumption, ensuring precise tracking and minimizing errors in daily reporting.
- ✓ **Enhanced Administrative Oversight**: Allows administrators to monitor meal plan usage, access comprehensive order histories, and generate detailed reports for better resource management.
- ✓ **User-Friendly Interface**: Provides an intuitive platform for students and staff, making meal selection, ordering, and management easy.
- ✓ **Scalability and Flexibility**: Designed to accommodate a growing student population while integrating seamlessly with existing campus infrastructures.

➢ "SCHOOLTHY: Automatic Menu Planner for Healthy and Balanced School Meals"

- ✓ **Dynamic Menu Display**: Generates and presents daily and weekly meal plans that are nutritionally balanced and visually appealing, keeping students engaged with their meal choices.
- ✓ **Nutritional Information Accessibility**: Displays detailed nutritional content, including calorie count, macronutrient breakdown, and allergen warnings, ensuring students and parents can make informed dietary decisions.
- ✓ **Customization for Dietary Needs**: Allows personalized meal selection, accommodating vegetarian, vegan, gluten-free, and other special dietary requirements.
- ✓ **User-Friendly Interface**: Features an intuitive design that makes it easy for students to view upcoming meals while enabling administrators to update and modify menus efficiently.
- ✓ **Integration with Educational Content**: Incorporates nutrition education to promote healthier food choices and improve student awareness of balanced diets.

1.6 References:

1. https://www.academia.edu/35633510/STUDENTS_MEAL_INFORMATION_SYSTEM
2. https://www.scribd.com/document/737703216/Meal-Management-system
3. https://www.academia.edu/106174493/A_Smart_Dietitian_using_BMR_Focused_on_Bangladeshi_Food_Habits
4. https://www.researchgate.net/publication/342249688_SCHOOLTHY_Automatic_Menu_Planner_for_Healthy_and_Balanced_School_Meals
5. https://www.delikopos.com/blog/meal-management-plan-for-students-through-student-cafeteria-management-system/

Chapter 2

Feasibility Study

2.1 Introduction:

Nutri-Track Academia is a planned website development project whose main focus is designing a "Students' Daily Meal Management System", preferably in a college/university setting. This project aims to create a fully functional website that will handle a student's daily meal planning and ordering activities. Functionality plans include generating a meal card for every individual student so that they can order their meals throughout their campus stay time using this card, options of calculating their BMI and BMR to suggest proper diet, proper meal menu display on the basis of time of the day, management of meal ordering and bill calculation, and so on. This website basically plans to be like a meal management assistant for a student, which will not only help them in purchasing meals at campus food outlets, but will also take care of their health by suggesting proper meal plans to each individual and displaying menus suited to each time of the day. The success of this project is believed to play an important role in ensuring the good health of students in their prime through proper meal management during a major part of their days.

2.2 Background:

Nutri-Track Academia is envisioned as a comprehensive solution to address the multifaceted challenges faced by students and campus dining facilities. Many students struggle to maintain a balanced diet due to their hectic academic schedules, limited budgets, and lack of awareness about proper nutrition. This often leads to irregular eating habits, dependence on unhealthy food options, and various health issues such as obesity, malnutrition, fatigue, and reduced cognitive performance—all of which can negatively impact academic success and overall well-being. To tackle these problems, the platform aims to create a fully functional meal planning and ordering system that streamlines students' dining experiences. It includes features like a digital meal card system for convenient ordering, BMI and BMR calculators for personalized dietary recommendations, and dynamic menus based on the time of day to help students make informed food choices. Additionally, the system facilitates seamless meal ordering, bill management, and order history tracking to improve efficiency and transparency. From an administrative perspective, it offers a dashboard for managing the entire meal system, allows menu customization, and maintains comprehensive records to ease workload and reduce errors in billing and order tracking. By digitizing and centralizing these processes, the platform not only promotes healthier eating habits and better student well-being but also enhances operational efficiency within campus dining services.

2.3 System Outline:

Nutri-Track Academia: A Students' Daily Meal Management System is a meal booking and billing system that manages student meal accounts for purchases of meals at the cafeteria on campuses. It's a planned series of events that is mainly concerned with menu planning, food purchasing, managing the bills, providing personalized dietary recommendations based on students' health metrics (BMI & BMR), allowing menu customization, and maintaining comprehensive order records for all students, ensuring transparency and tracking meal usage.

The complete outline of the project is described in Table 1:

Table 1: System Outline

Serial No.	**Item**	**Description**
01.	Users	1. Students 2. Cafeteria Managers 3. Administrator
02.	Existing Problems	1. Manual meal booking leads to errors. 2. Inefficient tracking of food consumption and wastage. 3. No real-time tracking of meal orders. 4. Payment handling is time-consuming. 5. Difficulty in meal customization for dietary preferences. 6. Unhealthy eating patterns among students
03.	Reason for Problems	1. Lack of an integrated digital system for meal and drink purchasing on campus. 2. A time-consuming process of placing an order through communication among students and cafeteria managers. 3. Limited availability of affordable and nutritious meal options on campus. 4. Lack of awareness regarding proper nutrition 5. No centralized platform for the admin to manage meal menus and order tracking.

04.	Performed Tasks	1. Fully functional meal planning and ordering platform. 2. Implement a digital meal card system. 3. Integrate BMI and BMR calculators to provide personalized dietary recommendations to students. 4. Display dynamic meal menus based on the time of day to help students make informed food choices. 5. Admin manages the entire meal system efficiently. 6. Inventory management for cafeteria staff.
05.	Required Data	1. Students' information. 2. Cafeteria Managers' information. 3. Admin Credentials 4. Nutritious Meal and Drinks information.

The logical model of the system is shown using DFD (Data Flow Diagram):

Admin DFD:

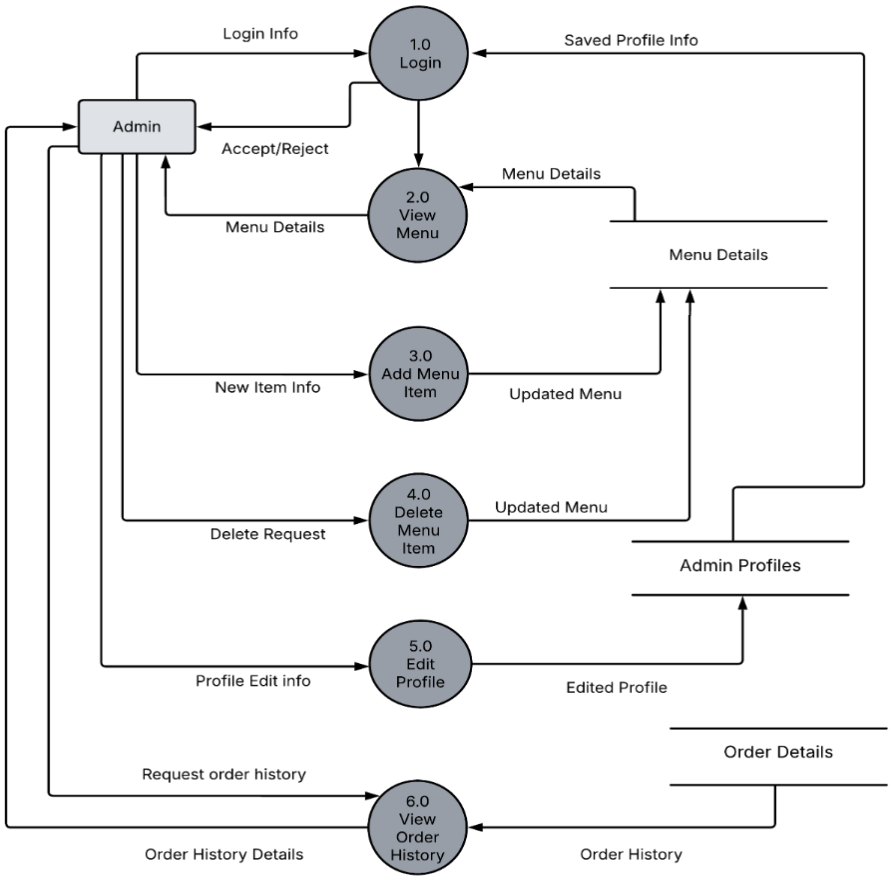


Figure 1: Admin DFD

Student DFD:

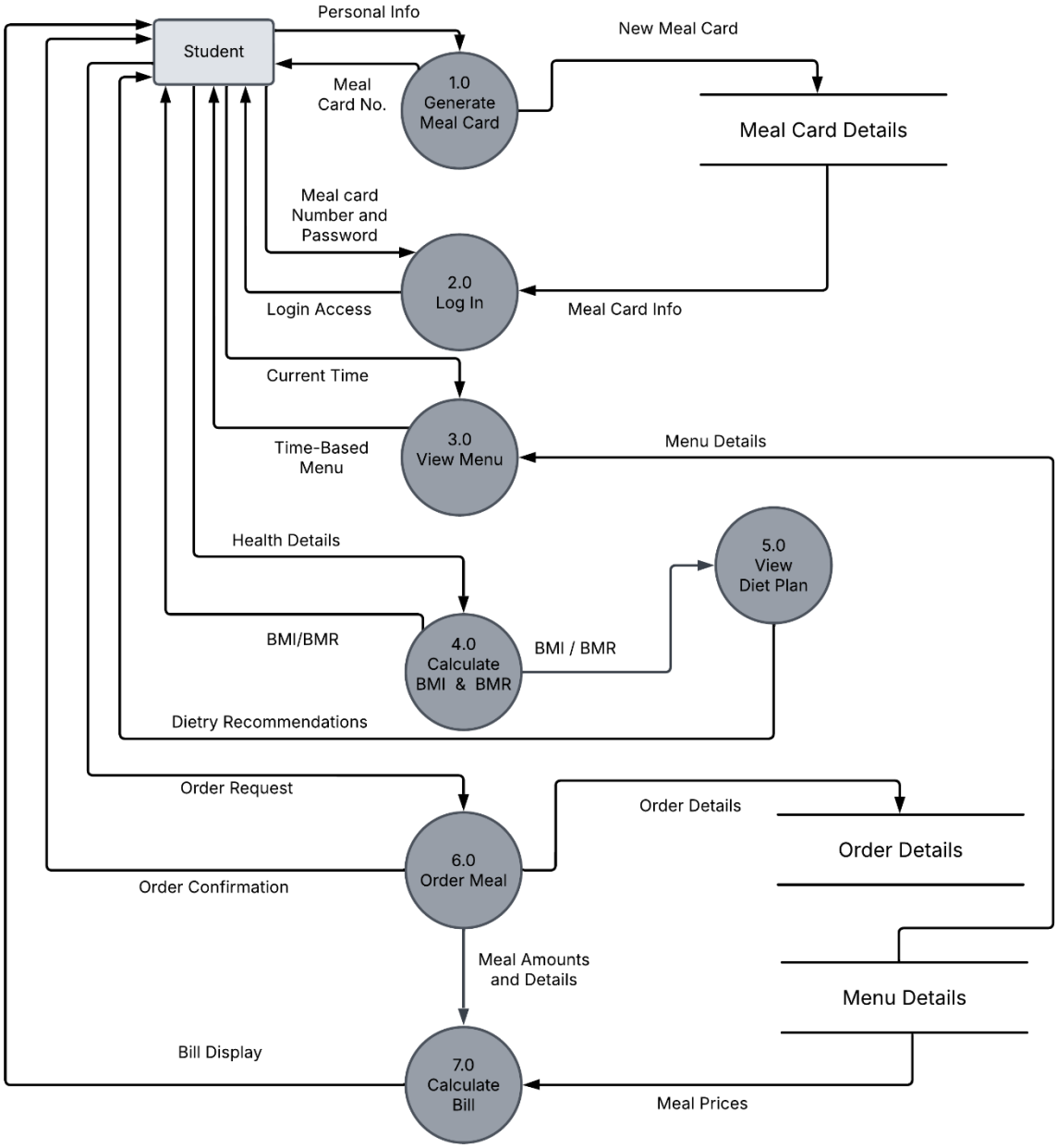


Figure 2: Student DFD

2.4 Methodology:

The methodology section outlines the systematic approach used to assess the feasibility and implementation strategy for developing a student daily meal management system. The goal is to ensure the platform is practical, efficient, and meets the needs of students, administrators, and campus dining services. The methodology includes the following key components:

i. **Requirements Gathering and Analysis:**
   - **User Needs Assessment:** Conduct surveys and interviews with students, campus staff, and canteen employees to understand their current challenges, preferences, and expectations regarding meal planning and ordering.
   - **Feature Benchmarking:** Analyze existing meal management systems (if any) to identify useful features, gaps, and opportunities for innovation specific to a student-centric environment.

ii. **Technical Feasibility:**
   - **Technology Stack Evaluation:** Determine the most suitable technologies for system development, including front-end frameworks (e.g., React), back-end systems (e.g., Django or Node.js), databases, and security protocols.
   - **System Architecture Design:** Develop a high-level architecture that includes modules for meal ordering, menu display, user authentication, billing, and admin control panels, ensuring scalability and security.

iii. **Operational Feasibility:**
   - **Process Flow Design:** Define the operational workflow of the system, such as meal card usage, real-time menu updates based on time slots, order tracking, and administrative controls for menu customization and record keeping.
   - **User Experience Planning:** Focus on intuitive design to ensure ease of use for students and staff, reducing learning curves and maximizing adoption.

iv. **Economic Feasibility:**
   - **Cost Estimation:** Analyze potential costs involved in development, deployment, maintenance, and training.
   - **Resource Optimization:** Explore strategies to minimize manual workload and operational costs for canteen staff by automating billing and record management.
   - **Efficiency Gains:** Assess potential benefits in terms of time saved, reduced billing errors, better inventory control, and improved health outcomes for students, which indirectly contribute to the value proposition of the system.

This structured methodology ensures a thorough evaluation of the student meal management system, enabling informed decision-making and guiding the project toward successful implementation and long-term sustainability.

2.5 Overview of Alternatives:

To improve meal management for students and overcome the inefficiencies of existing systems, several alternative solutions can be considered. The goal is to streamline the meal ordering process, reduce wait times, minimize manual workload, and offer a seamless payment experience. Each alternative has different levels of automation, efficiency, and cost-effectiveness.

The three main solutions for comparison are:

1. **Smart Table System (Hybrid Manual & Digital)** – A system where students order food through tablets provided at each canteen table. The tablets display a time-based menu, generate bills, and notify students when their food is ready. Payments are made through third-party apps like bKash, but this system requires a large dining area with multiple tablets, increasing costs.
2. **Third-Party Food Ordering App** – A fully digital system where students order meals via an external app. While it offers convenience, it lacks campus-specific features, relies on third-party payment systems with additional fees, and provides limited control for canteen admins.
3. **Students' Daily Meal Management System (Nutri-Track Academia)** – A fully automated web-based solution designed specifically for campus needs. It allows students to order food online, make payments through a prepaid meal card (eliminating third-party fees), track their order history, and access health features like BMI/BMR calculations and dietary recommendations. Unlike Smart Table, it doesn't require expensive tablets or extra dining space, making it more cost-effective and scalable.

The table below compares these alternatives based on key features to identify the most efficient and practical solution.

Table 2: Comparison among alternatives based on key features

Feature	**Smart Table System (Hybrid Manual & Digital)**	**Third-Party Food Ordering App**	**Students' Daily Meal Management System (Web-based System)**
Meal Ordering	Through tablets at canteen tables	Online via third-party app	Online via custom campus system
Menu Display	Tablets show time-based menu	App-based menu	Website shows time-based menu
Billing System	Bill calculated on tablet, paid via third-party app (e.g., bKash)	Automated, but includes transaction fees	Automated, integrated with student meal card
Payment Method	Third-party app (extra charges apply)	Third-party app (extra charges apply)	Prepaid meal card (no extra fees)
Health Tracking (BMI/BMR, Diet Recommendations)	Not available	Not available	Available

Student Meal Card Integration	No	No	Yes
Order Notifications	Tablet notifies when food is ready	App notifies order status	Website notifies order status
Order History Tracking	Available on tablet	Not available for canteen admins	Available for admins & students
Queue & Waiting Time	Moderate (students wait for tables & tablets)	Varies (depends on delivery/canteen pickup)	Minimal (order from anywhere, pick up when ready)
Device Dependency	Requires tablets at every table (expensive)	Requires smartphone & internet	Works on any device, no extra hardware needed
Customization for Campus Needs	Limited (relies on canteen setup)	Limited (dependent on third-party platform)	Fully customizable
Admin Controls (Menu & Order Management)	Can update menu, but limited reporting	Limited (third-party controls order system)	Full control over menu and orders
Automation & Efficiency	Partial (still requires staff monitoring)	Moderate (Automated, but commission-based)	High (Fully automated & optimized for campus)
Scalability	Limited (more tablets needed as students increase)	High (Third-party handles scaling)	High (Can expand features as needed)
Land & Area Requirements	High – More space needed for tables, tablets, and manual canteen setup	Low – Mostly relies on delivery/pickup system	Low – No need for extra space beyond the canteen itself

2.5.1 Economic Feasibility Study for the Alternatives:

1. Smart Table System (Hybrid Manual & Digital Approach)

Description:

- A mix of manual and digital processes where students order meals through tablets at canteen tables.
- Tablets display time-based menus, generate bills, and notify students when meals are ready.
- Payments are made through third-party apps like bKash, leading to extra charges.
- Requires a large dining area with multiple tablets, increasing operational costs.

Table 3: Estimated Costs (Per Year):

Cost Component	Estimated Cost (BDT)
Tablets for Canteen Tables (200 Tablets @ 30,000 BDT Each, One-Time)	60,00,000
Tablet Maintenance & Replacement (20% Yearly)	12,00,000
Payment Gateway & Transaction Fees	5,00,000
Employee Salaries (Meal Distribution & Order Processing)	10,00,000
Additional Space & Infrastructure for Tablet-Based Dining	8,00,000
Total Cost (First Year)	95,00,000
Total Cost (From Second Year)	35,00,000

2. Third-Party Food Ordering App (Outsourced Solution)

Description:

- A pre-built system where students order meals via a third-party app.
- Requires integration with external platforms, leading to high transaction fees.
- Limited customization options for the campus canteen.
- No dedicated meal card system, requiring reliance on external payment services.

Table 4: Estimated Costs (Per Year):

Cost Component	Estimated Cost (BDT)
Software Subscription Fee	14,00,000
Integration & Setup (One-Time)	5,50,000
Payment Gateway & Transaction Fees	3,50,000
Data Security Risks & External Hosting	2,50,000
Employee Salaries (Meal Distribution & Admin Work)	10,00,000
Total Cost (First Year)	35,50,000
Total Cost (From Second Year)	30,00,000

3. Nutri-Track Academia: Students' Daily Meal Management System (Web-Based System)

Description:

- A custom web-based platform designed specifically for the campus canteen.
- Students can order meals online, use a prepaid meal card for cashless transactions, and track order history.
- Eliminates the need for expensive tablets and third-party payment fees.
- Provides extra health features like BMI/BMR calculations and dietary recommendations.
- Reduces queue times and minimizes employee workload.

Table 5: Estimated Costs (Per Year):

Cost Component	**Estimated Cost (BDT)**
Website Development (One-Time)	12,00,000
Hosting & Maintenance (Per Year)	2,50,000
Security & Database Management (Per Year)	1,80,000
Employee Training for System Use (One-Time)	1,20,000
Employee Salaries (Meal Distribution & Admin Work)	9,00,000
Total Cost (First Year)	26,50,000
Total Cost (From Second Year)	13,30,000

Financial Analysis for Alternative 1 (Smart Table):

Table 6: Benefit vs. Cost Analysis

Benefits			**Costs**		
Serial No.	**Particular**	**Amount (BDT)**	**Serial No.**	**Particular**	**Amount (BDT)**
1	Increased Efficiency in Ordering	15,00,000	1	Tablets for Canteen Tables (200 @ 30,000 BDT Each, One-Time)	60,00,000
2	Reduced Queue Time	10,00,000	2	Tablet Maintenance & Replacement (20% Yearly)	12,00,000
3	Improved Customer Satisfaction	12,00,000	3	Payment Gateway & Transaction Fees	5,00,000
4	Scalability for Large Student Base	8,00,000	4	Employee Salaries (Meal Distribution & Order Processing)	10,00,000

5	Automated Billing System	5,00,000	5	Additional Space & Infrastructure for Tablet-Based Dining	8,00,000
Total Benefits		**50,00,000**	**Total Costs (1st year)**		**95,00,000**
			Total Costs (From 2nd year)		**35,00,000**
Net Return per Year		**50,00,000-35,00,000 =15,00,000**			

Table 7: Investment Analysis

Year	Savings (BDT)	Present Value (BDT) (13%)	Cumulative Value (BDT)
1	15,00,000	13,27,433	13,27,433
2	15,00,000	11,74,720	25,02,153
3	15,00,000	10,39,575	35,41,728
4	15,00,000	9,19,978	44,61,706
5	15,00,000	8,14,140	52,75,846
6	15,00,000	7,20,478	59,96,324
s7	15,00,000	6,37,591	66,33,915
8	15,00,000	5,64,240	71,98,155
9	15,00,000	4,99,327	76,97,482
10	15,00,000	4,41,883	81,39,365

Financial Analysis for Alternative 2 (Third-Party Food Ordering App):

Table 8: Benefit vs. Cost Analysis

Benefits			**Costs**		
Serial No.	**Particular**	**Amount (BDT)**	**Serial No.**	**Particular**	**Amount (BDT)**
1	Quick Deployment	12,00,000	1	Software Subscription Fee	14,00,000
2	No Hardware Maintenance	10,00,000	2	Integration & Setup (One-Time)	5,50,000
3	Increased Scalability	8,00,000	3	Payment Gateway & Transaction Fees	3,50,000

4	Reduced Manual Errors	7,00,000	4	Data Security Risks & External Hosting	2,50,000
5	Improved Customer Convenience	6,00,000	5	Employee Salaries (Meal Distribution & Admin Work)	10,00,000
Total Benefits		**43,00,000**	**Total Costs (1st year)**		**35,50,000**
			Total Costs (From 2nd year)		**30,00,000**
Net Return per Year		**43,00,000-30,00,000 =13,00,000**			

Table 9: Investment Analysis

Year	Savings (BDT)	Present Value (BDT) (13%)	Cumulative Value (BDT)
1	13,00,000	11,50,442	11,50,442
2	13,00,000	10,18,091	21,68,533
3	13,00,000	9,00,965	30,69,498
4	13,00,000	7,97,314	38,66,812
5	13,00,000	7,05,588	45,72,400
6	13,00,000	6,24,414	51,96,814
7	13,00,000	5,52,579	57,49,393
8	13,00,000	4,89,008	62,38,401
9	13,00,000	4,32,750	66,71,151
10	13,00,000	3,82,965	70,54,116

Financial Analysis for Alternative 3 (Nutri-Track Academia: Students' Daily Meal Management System):

Table 10: Benefit vs. Cost Analysis

Benefits			**Costs**		
Serial No.	**Particular**	**Amount (BDT)**	**Serial No.**	**Particular**	**Amount (BDT)**
1	Automated Meal Ordering	18,00,000	1	Website Development (One-Time)	12,00,000

2	Elimination of Third-Party Fees	12,00,000	2	Hosting & Maintenance (Per Year)	2,50,000
3	Reduced Employee Workload	10,00,000	3	Security & Database Management (Per Year)	1,80,000
4	Faster Service & Convenience	8,00,000	4	Employee Training for System Use (One-Time)	1,20,000
5	Health-Focused Features (BMI/BMR)	7,00,000	5	Employee Salaries (Meal Distribution & Admin Work)	9,00,000
Total Benefits		**55,00,000**	**Total Costs (1st year)**		**26,50,000**
			Total Costs (From 2nd year)		**13,30,000**
Net Return per Year		**55,00,000-13,30,000 =41,70,000**			

Table 11: Investment Analysis

Year	Savings (BDT)	Present Value (BDT) (13%)	Cumulative Value (BDT)
1	41,70,000	36,90,265	36,90,265
2	41,70,000	32,65,722	69,55,987
3	41,70,000	28,90,019	98,46,006
4	41,70,000	25,57,539	1,24,03,545
5	41,70,000	22,63,309	1,46,66,854
6	41,70,000	20,02,928	1,66,69,782
7	41,70,000	17,72,503	1,84,42,285
8	41,70,000	15,68,587	2,00,10,872
9	41,70,000	13,88,130	2,13,99,002
10	41,70,000	12,28,433	2,26,27,435

2.5.2 Technical Feasibility Study of Alternatives

1. **Smart Table System (Hybrid Manual & Digital)** – A system where students order food through **tablets** installed at each canteen table. The tablets display a time-based menu, generate bills, and notify students when food is ready. Payments are made through third-party apps like bKash.
2. **Third-Party Food Ordering App** – A fully digital system where students use an **external** app to order meals. The system provides convenience but lacks campus-specific features, relies on third-party payment systems, and incurs additional fees.
3. **Students' Daily Meal Management System (Proposed System)** – A **fully automated web-based solution** built specifically for university needs. Students can order food online, make payments via a **prepaid meal card**, track orders, and access health-related features.

Table 12: **Technical Feasibility Analysis**

Criteria	Smart Table System	Third-Party App	Nutri-Track Academia
Complexity	High – Hardware integration needed.	Medium – Uses existing infrastructure.	Medium – Web development required.
Infrastructure	High – Tablets, WiFi, server.	Moderate – External Cloud Services	Low –Only web server and database.
Payment System	Third-party apps, transaction fees.	Third-party gateways, added costs.	Prepaid meal card, lower fees.

2.5.3 Operational Feasibility Study of Alternatives:

1. **Smart Table System** is **operationally challenging** due to **high hardware costs**, **staff workload**, and **scalability issues**.
2. **Third-party food Ordering App** is **convenient** but **limits university control** and adds **ongoing transaction fees**.
3. **Students' Daily Meal Management System (Proposed System)** is the most **operationally feasible**, as it requires **minimal staff involvement**, **low maintenance**, **cost-effective payments**, and **high scalability** for future improvements.

Table 13: **Operational Feasibility Analysis**

Criteria	Smart Table System	Third-Party App	Nutri-Track Academia
Cost of Implementation	High – Expensive tablets, software, and maintenance.	Low – No setup cost, but transaction fees apply.	Moderate – Development & hosting costs, but cost-effective long-term.
User Convenience	Moderate – Orders placed only at tables.	High – Mobile ordering from anywhere.	High – Web-based, no wait times, order tracking.

Canteen Workload	High – Staff must manage tablets and connectivity.	Moderate – Orders automated, but third-party dependent.	Low – Direct kitchen orders, reducing manual work.
Payment System	Moderate – Third-party apps, manual verification.	Low – External handling with processing fees.	High – Prepaid meal card, faster and fee-free.
Maintenance & Viability	High – Frequent maintenance, high costs.	Moderate – External dependency for updates.	Low – Web-based, easy updates, managed internally.

2.6 Recommendation:

After a comprehensive analysis of all alternatives, the **Students' Daily Meal Management System (Nutri-Track Academia)** is strongly recommended as the most **technically**, **operationally**, and **economically feasible** solution for campus meal management. Unlike the **Smart Table System**, which requires high-cost hardware and maintenance, or **Third-Party Food Ordering Apps**, which reduce institutional control and increase transaction costs, Nutri-Track Academia offers a **cost-effective**, **campus-specific**, and **scalable** digital platform.

Table 14: Comparison among Alternatives based on economic feasibility study

SL.NO.	Feature	Alternative 1	Alternative 2	Alternative 3
01.	Initial Development Cost	95,00,000	35,50,500	26,50,000
02.	Recurring Costs (Per Year)	35.00,000	30,00,000	13,30,000
03.	Net Return (Per Year)	15,00,000	13,00,000	41,70,000
04.	Present Value (13% Discount Rate)	4,41,883	3,82,965	12,28,433
05.	Cumulative Value (10 Years)	81,39,365	70,54,116	2,26,27,435

06.	System Life Cycle	10 years	10 years	10 years
07	Return Value	00,00,000	38,66,812	36,90,265
08.	Payback Period	>10 years	4 years	1 year

Hence, this system ensures automated ordering**,** personalized meal recommendations**,** reduced staff workload**,** and streamlined billing through a prepaid meal card system, eliminating third-party fees. It enhances student convenience, supports nutritional awareness, and simplifies cafeteria operations, making it a sustainable and future-ready solution for campus meal management.

2.7 Conclusion:

Nutri-Track Academia: Students' Daily Meal Management System (Web-Based System) stands out as the most viable solution due to its automation, cost efficiency, and scalability. Unlike other alternatives, it streamlines meal ordering, enhances nutritional tracking, and minimizes administrative effort. By integrating a digital meal card and personalized dietary recommendations, it improves user experience while reducing operational complexities. This approach ensures a more efficient, transparent, and future-ready meal management system for university campuses.

Chapter 3

Requirement Specification

3.1 Introduction:

The Software Requirement Specification (SRS) is the official statement of what is required of the system developers. It should include both a definition of user requirements and a specification of the system requirements. It is a description of a software system to be developed. It lays out functional or non-functional requirements and includes a set of use cases that describe user interactions that the software must provide.SRS establishes the basis for an agreement between the customer and contractors or suppliers on what the software product is to do, as well as what it is not expected to do.

Nutri-Track Academia is a planned website development project whose main focus is designing a "Students' Daily Meal Management System", preferably in a college/university setting, aiming to create a fully functional website that will handle a student's daily meal planning and ordering activities. The system will provide real-time insights to the university administration and dining staff to optimize food preparation and reduce waste. It will include features such as students' profiles, dietary recommendations, order management, and canteen staff dashboards.

This document provides a detailed description of the software functionalities, design constraints, user interface, performance requirements, and dependencies of the **Nutri-Track Academia** system. It forms the basis for the development and validation of the application to ensure it meets user and institutional needs effectively.

3.2 User Requirement Specifications:

3.2.1 Functional Requirements:

The functional requirements describe what a system should do – the specific features it must contain to satisfy user needs. The functional requirements of **Nutri Track Academia – A Students' Daily Meal Management System** are:

1. User Registration and Authentication

- The system shall allow students to register with name, email, ID, university PIN and password.
- The system shall generate a meal card for the registered student using their ID and university PIN.
- The system shall allow students to log in using their credentials.
- The system shall validate login credentials and deny access if incorrect.

2. Meal Card Management

- The system shall associate each student with one unique meal card.
- The meal card shall be used for logging in and ordering meals.

3. Menu Management

- The system shall categorize menus into Breakfast, Lunch, Snacks, and Drinks.
- The system shall allow admins to add, update, and delete menu items.
- The system shall allow students to view available menu items by category and current time of the day.

4. Order Processing

- The system shall allow students to place food orders using their meal card.
- The system shall store order details (item, quantity, timestamp).
- The system shall allow students to view their past order history.

5. Billing System

- The system shall automatically generate a bill after an order is placed.
- The system shall calculate the total price based on selected items.
- The system shall show bill details (amount, date, items).

6. BMI/BMR & Diet Chart

- The system shall allow students to enter height, weight, and age.
- The system shall calculate BMI and BMR based on the input.
- The system shall generate a diet chart and provide dietary advice.

7. Admin Functionalities

- The system shall allow admins to log in securely.
- The system shall allow admins to edit their own profile.
- The system shall allow admins to manage menus (update item details, add new items, delete items).
- The system shall allow admins to view students' BMI/BMR for health tracking.
- The system shall allow admins to view students' order history in order to manage inventory based on them.

3.2.2 Non-Functional Requirements:

Non-Functional Requirements are not directly concerned with the specific services delivered by the system to its users. It may affect the overall architecture of a system rather than the individual components. A single non-functional requirement may generate a number of related functional requirements. The non-functional requirements of **Nutri Track Academia – A Students' Daily Meal Management System** are:

1. **Performance**
   - The system must support at least 100 concurrent users (admins, cafeteria staff, and health staff) without lag.
   - All core operations (e.g., viewing meal menus, loading student BMI/BMR charts, generating order history) must respond within 2–3 seconds.
   - Data updates (e.g., adding a new meal or updating a profile) should reflect in real-time.

2. **Security**
   - User authentication is required for admin login using encrypted credentials.
   - Sensitive students' data (health metrics like BMI/BMR, order history) must be stored and transmitted using AES or SSL encryption.
   - The system should be protected against SQL injection, XSS, and unauthorized access attempts.
3. **Scalability**
   - The system must be designed to scale for future growth (e.g., if additional campus or more students are added).
   - Meal, order, and health data should remain efficient even with thousands of records.
4. **Reliability**
   - The system must have 99.9% uptime during campus hours (critical operational window).
   - Automatic daily backups of all students, meal, and order data must be maintained.
   - The system must recover from failure within 2 minutes to ensure continuity.
5. **Usability**
   - The interface must be intuitive and simple for non-technical users such as cafeteria staff.
   - Use of clear labels, menu icons, and guided workflows for common tasks (e.g., editing meals, viewing reports).
   - The system should be accessible with minimal training—less than 1 hour for admin-level users.
6. **Maintainability**
   - The system architecture should be modular and layered, allowing for easy updates or upgrades.
   - Source code should be well-documented and follow consistent naming and structure standards.
7. **Compatibility**
   - The system must run smoothly across all major browsers (Chrome, Firefox, Edge) and devices (desktop, tablet, mobile).
   - Should be responsive and mobile-friendly for cafeteria and on-the-go staff access.
8. **Compliance**
   - Adhere to local educational and health data regulations related to storing students' medical and nutritional data.

- All health-related information must be handled with parental or guardian consent where applicable.

3.3 Stakeholders of the System:

The term stakeholder is used to refer to any person or group who will be affected by the system, directly or indirectly. Stakeholders interact with the system by receiving services from it & by providing data & controlling signals to the system.

Stakeholders can be categorized as:

1. Primary stakeholder- is one who will be the end user of the proposed system and therefore, will receive the services functions provided by the system and will interact with the system interface.

2. Secondary stakeholder- on the other hand, is one who requires information from the system but is not necessarily going to use the system directly.

3. Ternary stakeholder- users will benefit from the system in an implicit way

Identifying and addressing the needs and requirements of these stakeholders is crucial for the successful development and operation of the Students' Daily Meal Management System. It helps tailor user interfaces and roles, prioritize features (like order tracking for students or nutritional analytics for admins), design better workflows and feedback mechanisms. The stakeholders of **Nutri-Track Academia** are:

I. Primary Stakeholders

Primary Stakeholder	Role/Interest
Students	End users; they place orders, track meals, give feedback, and manage preferences.
University Administration	Oversees system operations, monitors usage, allocates budget, and ensures efficiency.
Canteen Staff	Prepares and serves meals based on the system's daily requirements and orders.
System Admins/IT Team	Maintains and manages the software, hardware, updates, and security.

II. Secondary Stakeholders

Secondary Stakeholder	Role/Interest
Finance/Accounts Department	Handles billing, payment integration, subsidies, and budgeting.
Healthcare Unit	Could use the system's dietary data to support student wellness.

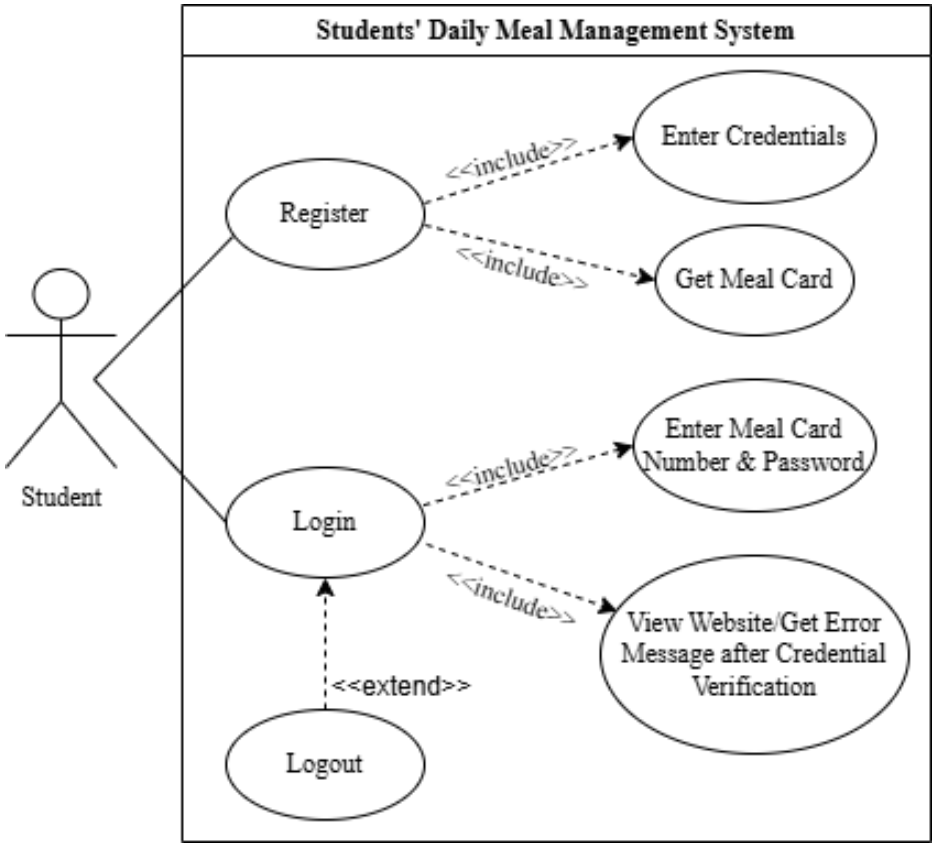
Policy Makers	Use data analytics to improve student services, nutrition, or policy formulation.

III. Ternary Stakeholders

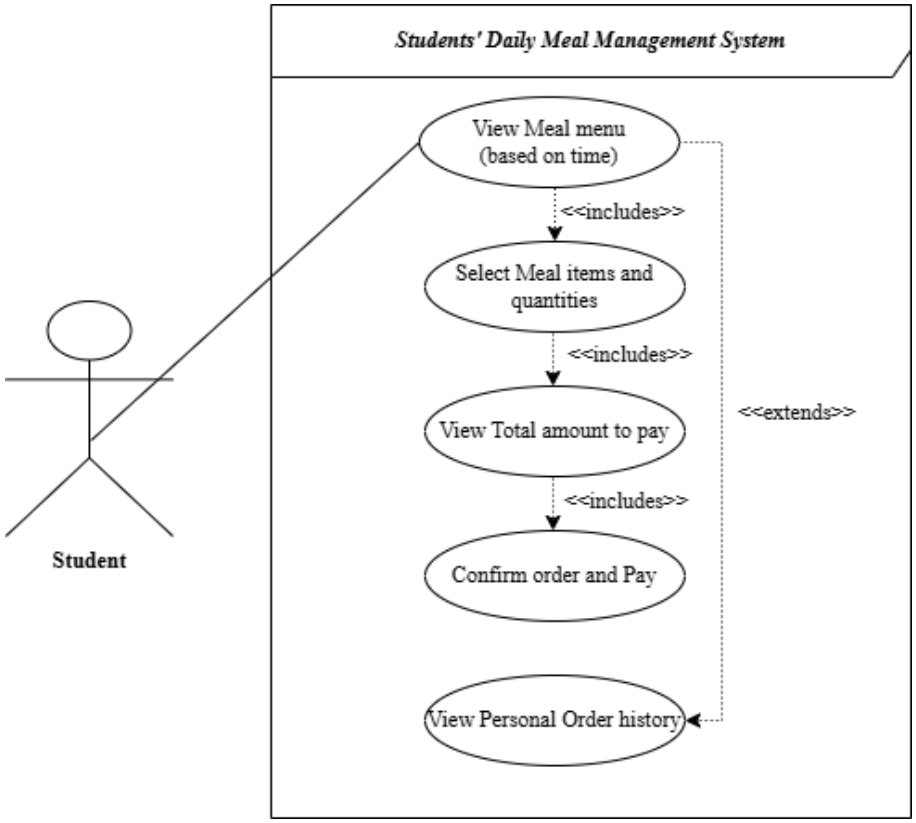
Ternary Stakeholder	Role / Relevance
Mobile App Platforms	App stores and services used to distribute or support the mobile version of the system.
Cybersecurity Auditors	External evaluators ensuring student data and transactions remain secure.
Educational Partners/Researchers	Use anonymized meal and nutrition data for research or academic purposes.

3.4 Use Case Diagrams:

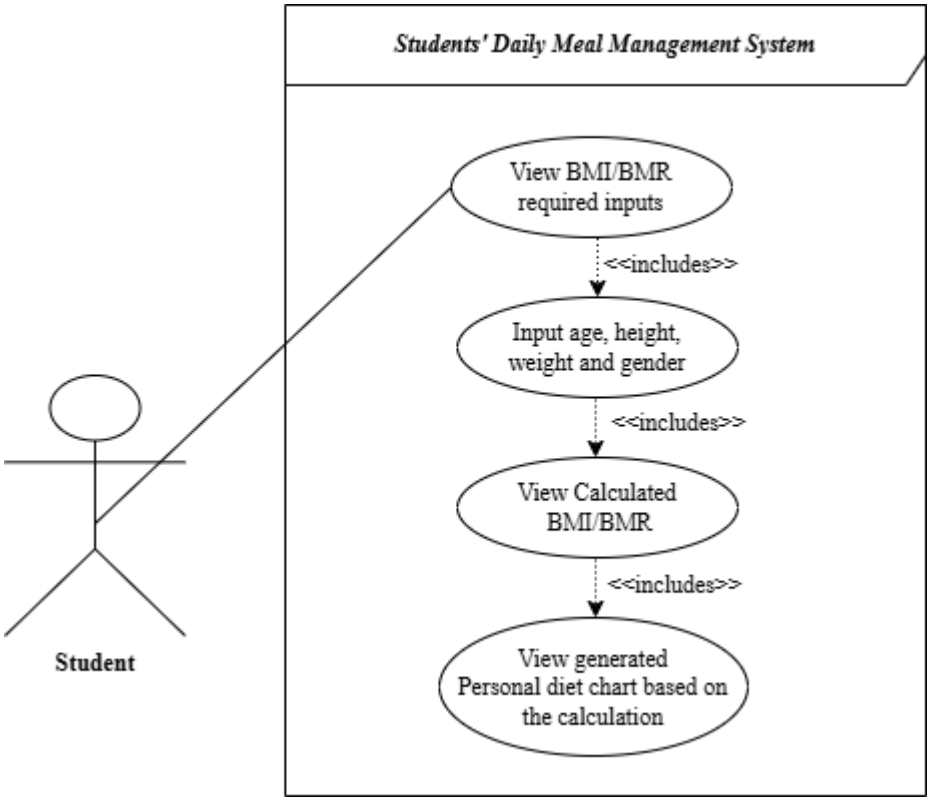
3.4.1 Student Register/Login:



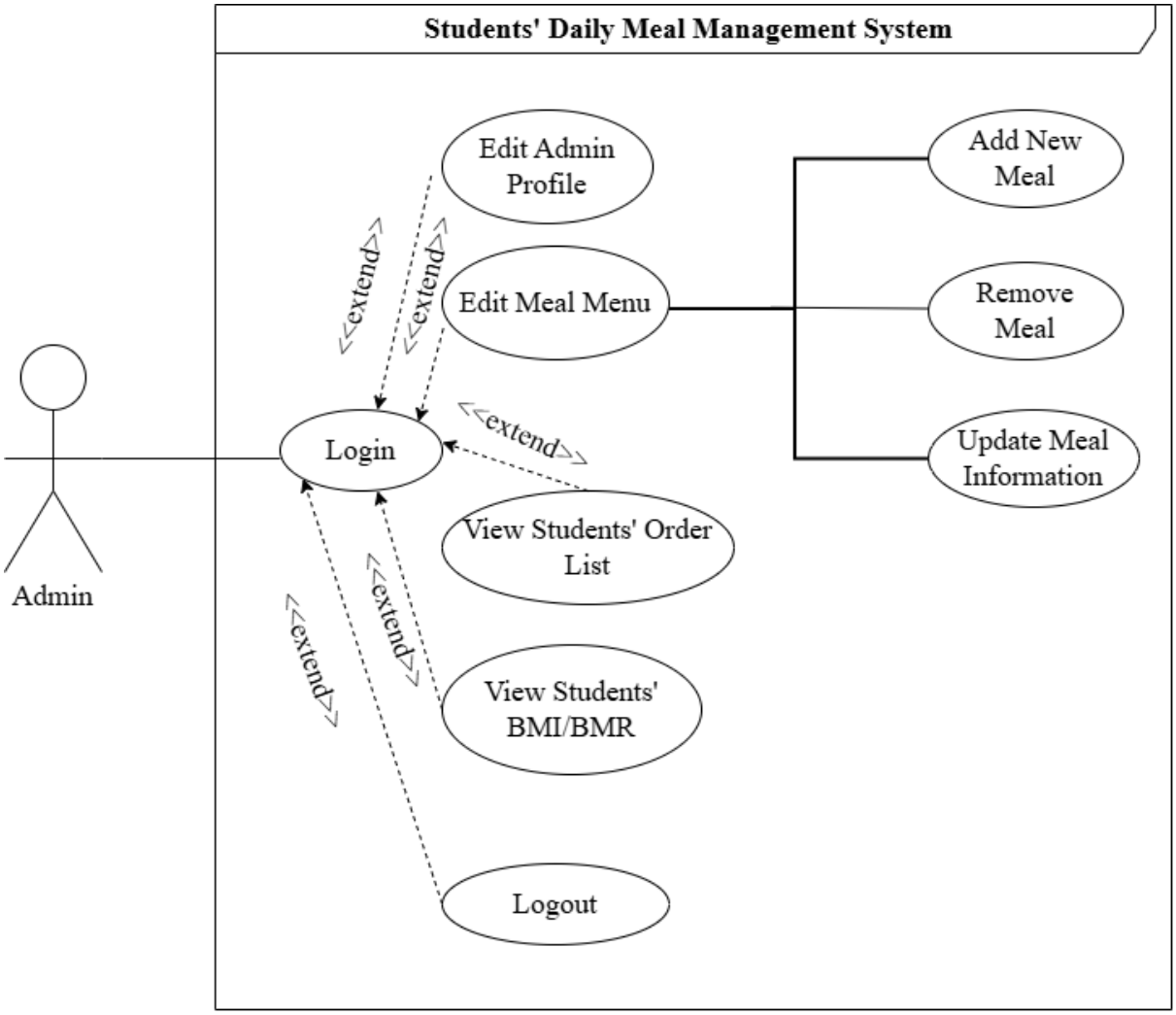
3.4.2 Meal-Menu Ordering Properties:



3.4.3 BMI/BMR Calculation Properties:

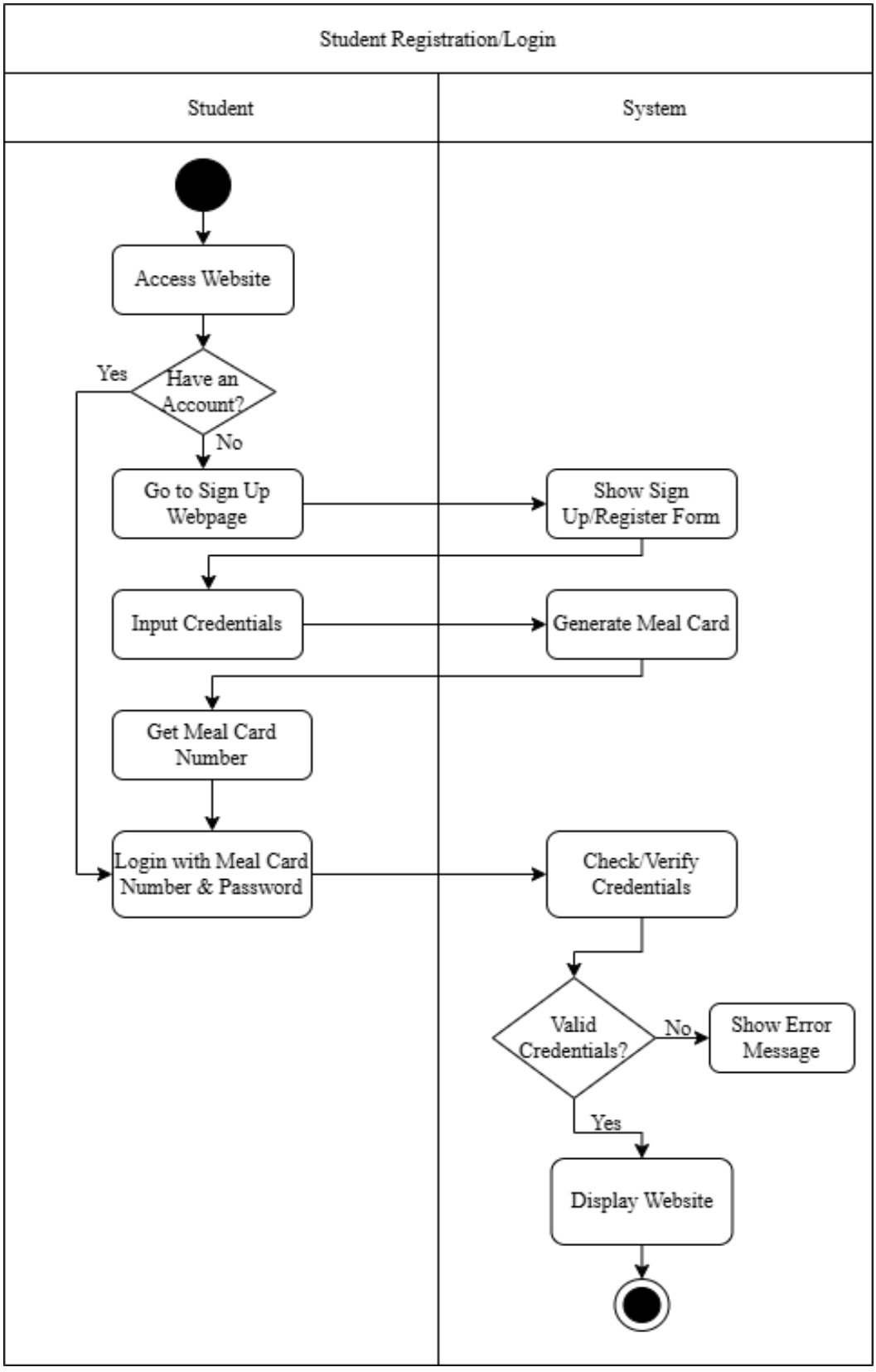


3.4.4 Admins' Functionalities:

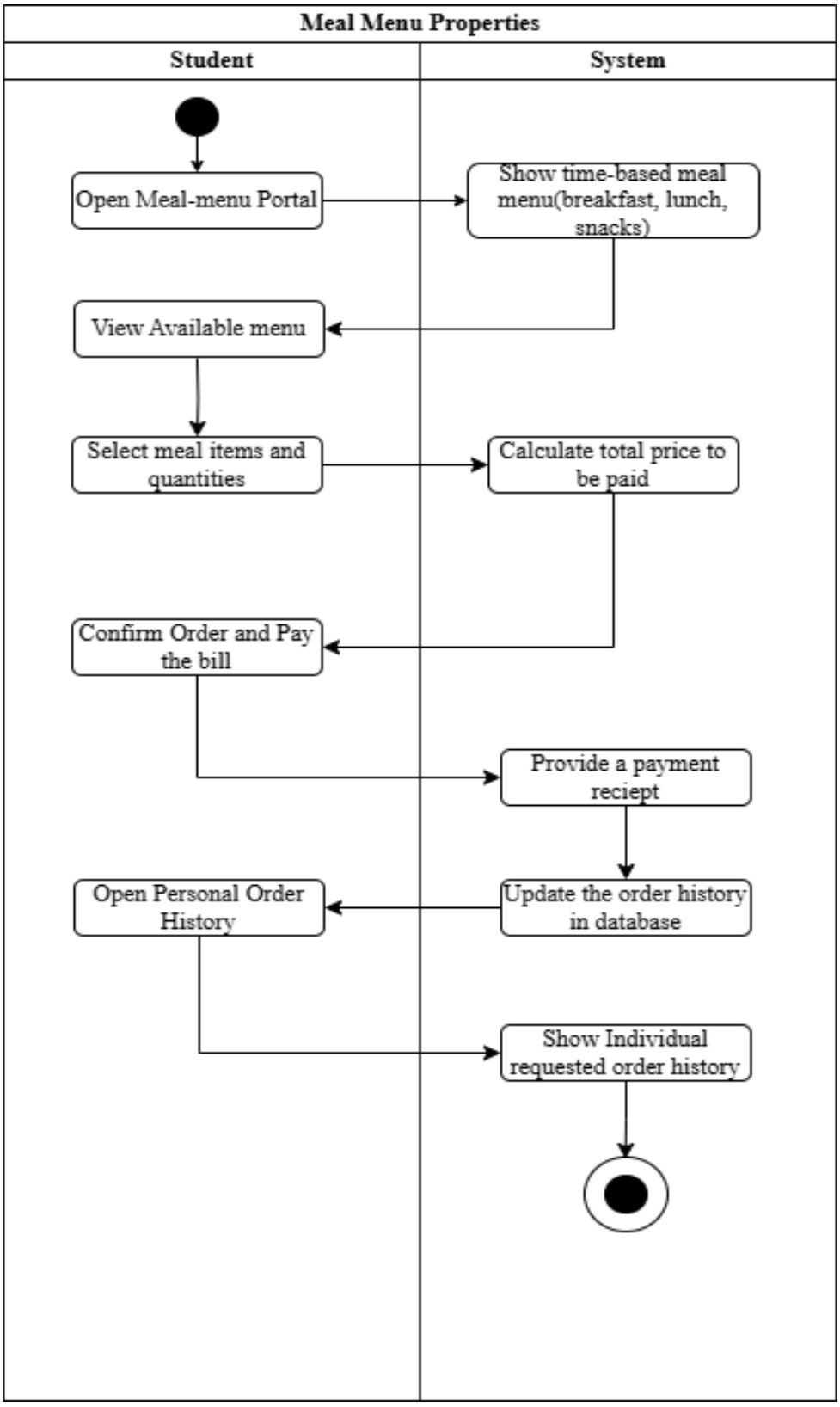


3.5 Activity Diagrams:

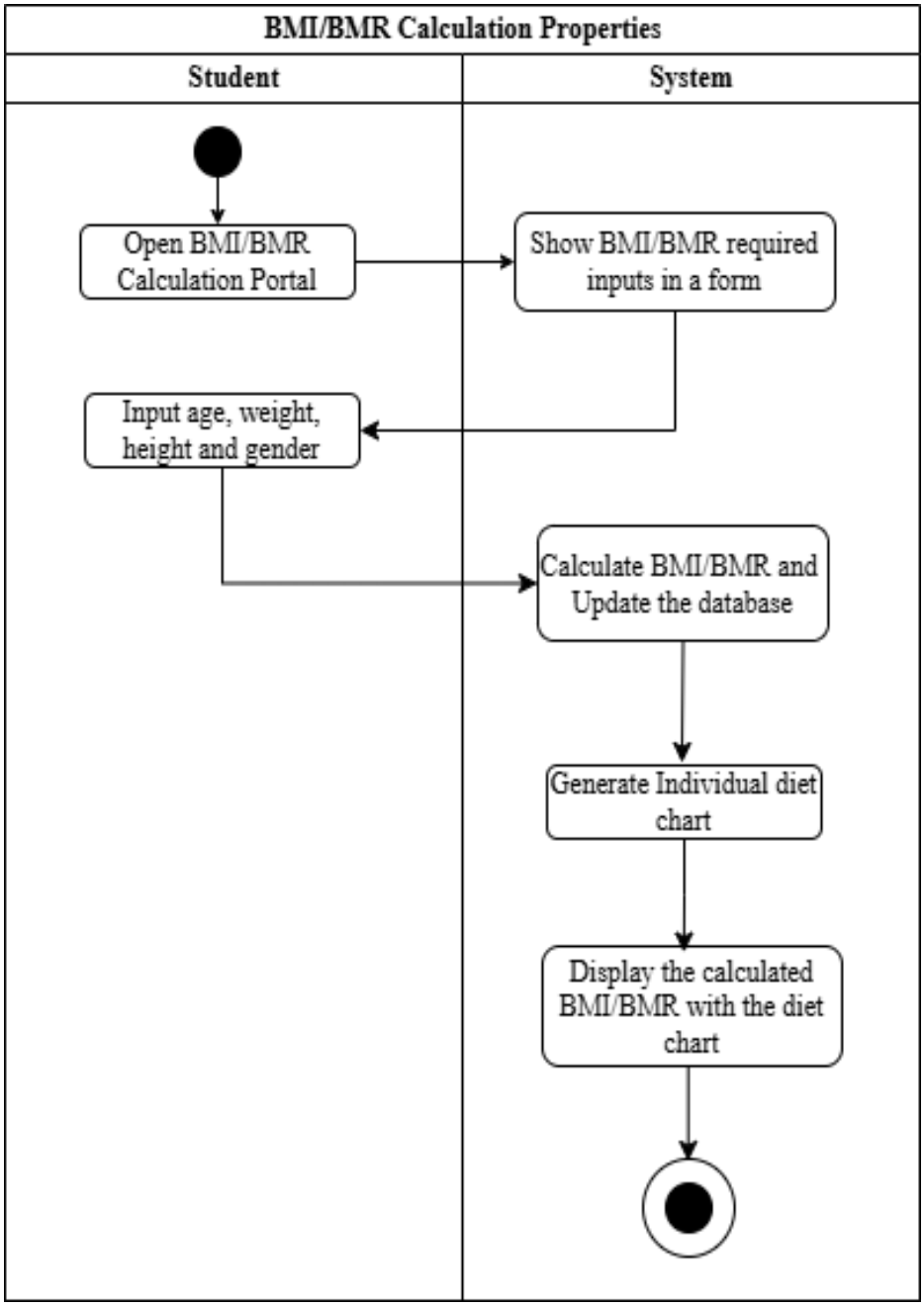
3.5.1 Student Register/Login:



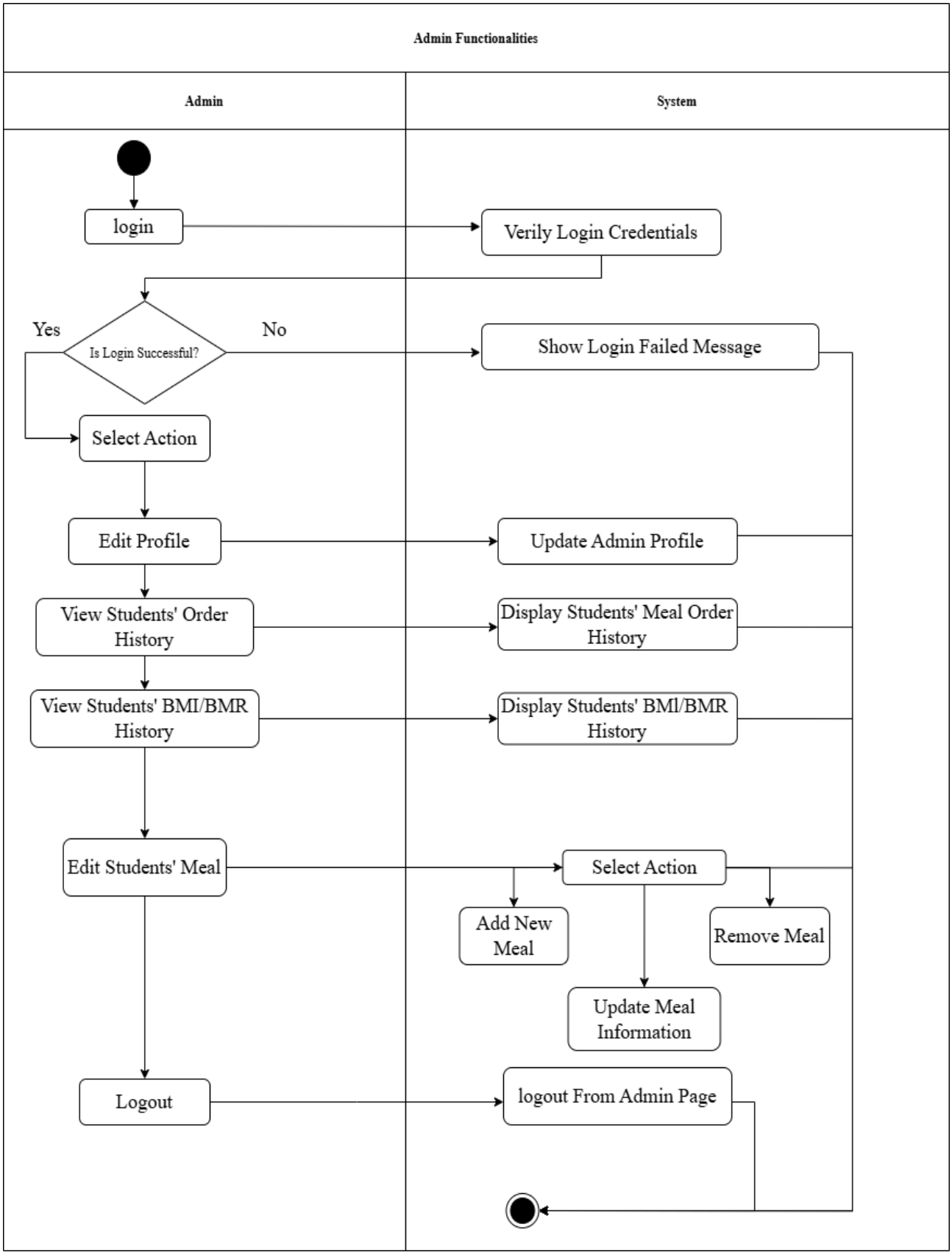
3.5.2 Meal-Menu Properties:



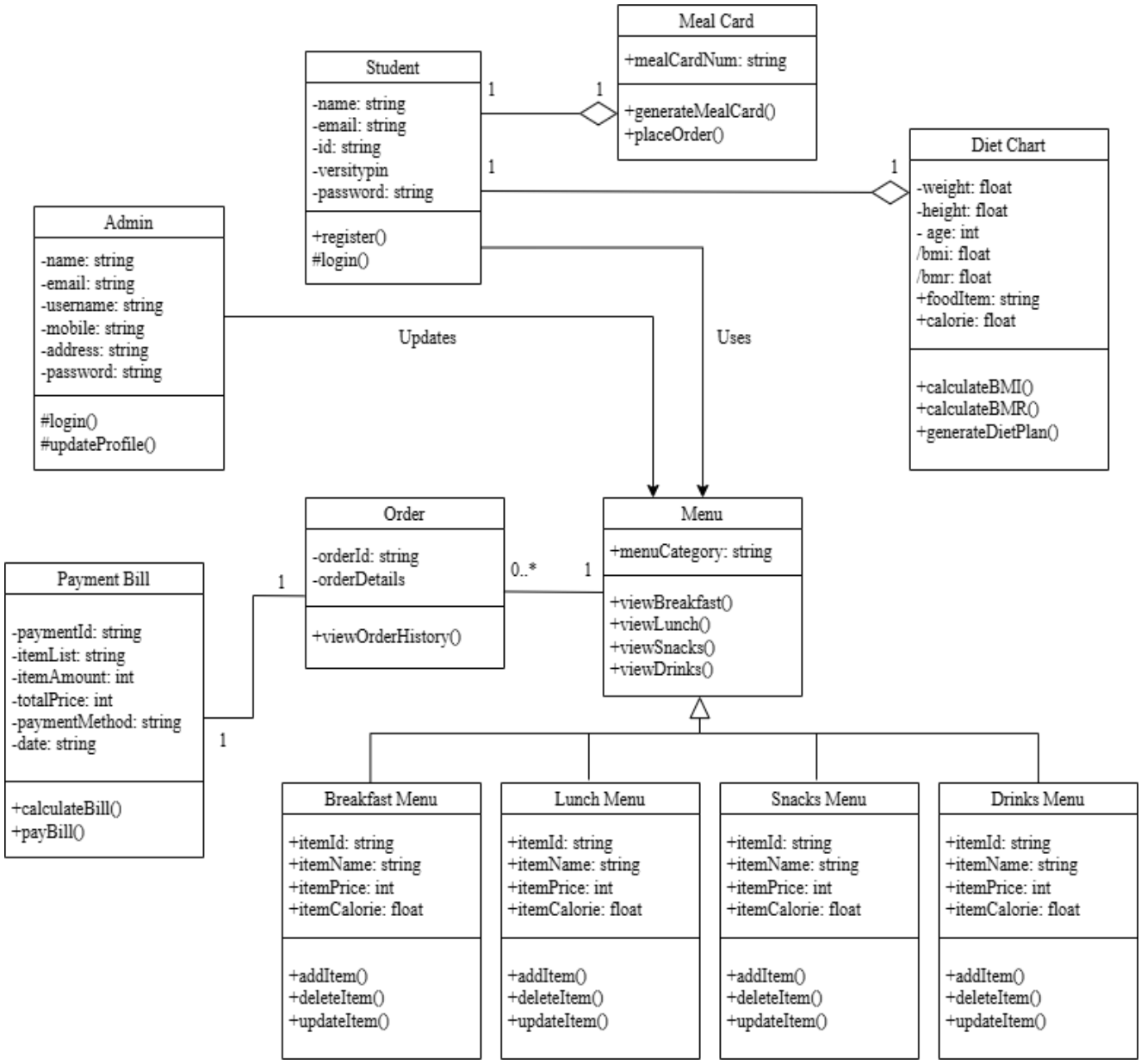
3.5.3 BMI/BMR Calculation properties:



3.5.4 Admins' Functionalities:

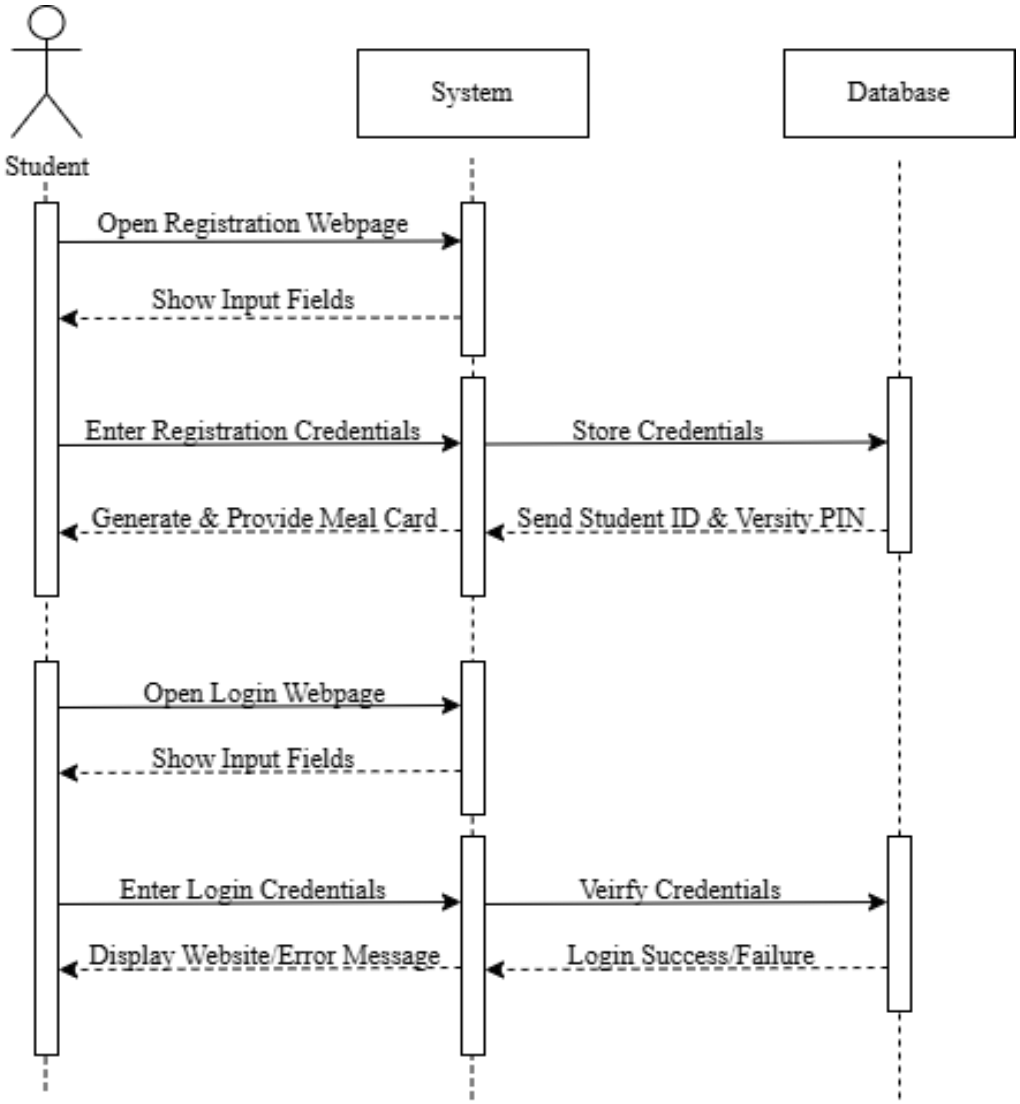


3.6 Class Diagram:

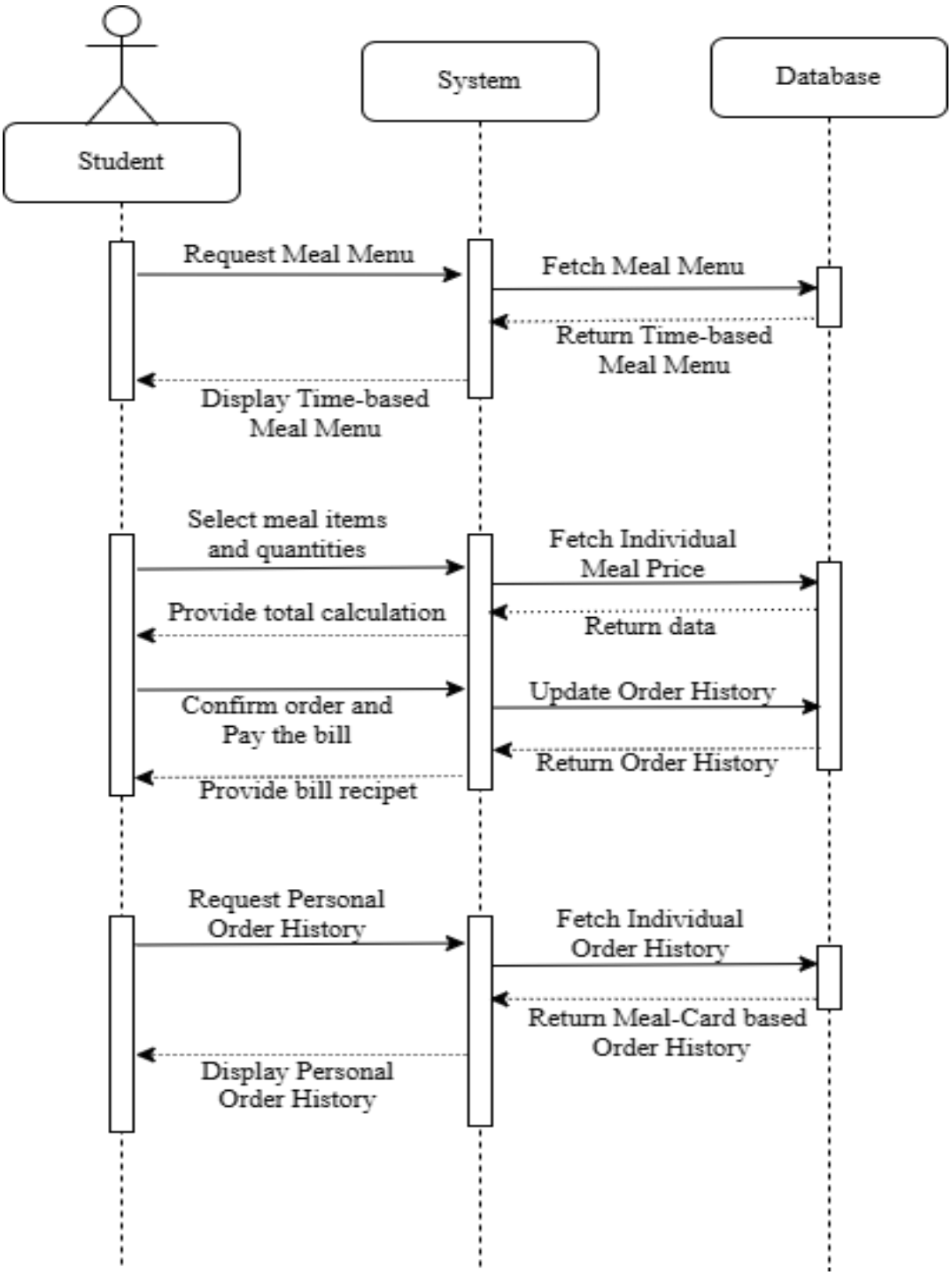


3.7 Sequence Diagrams:

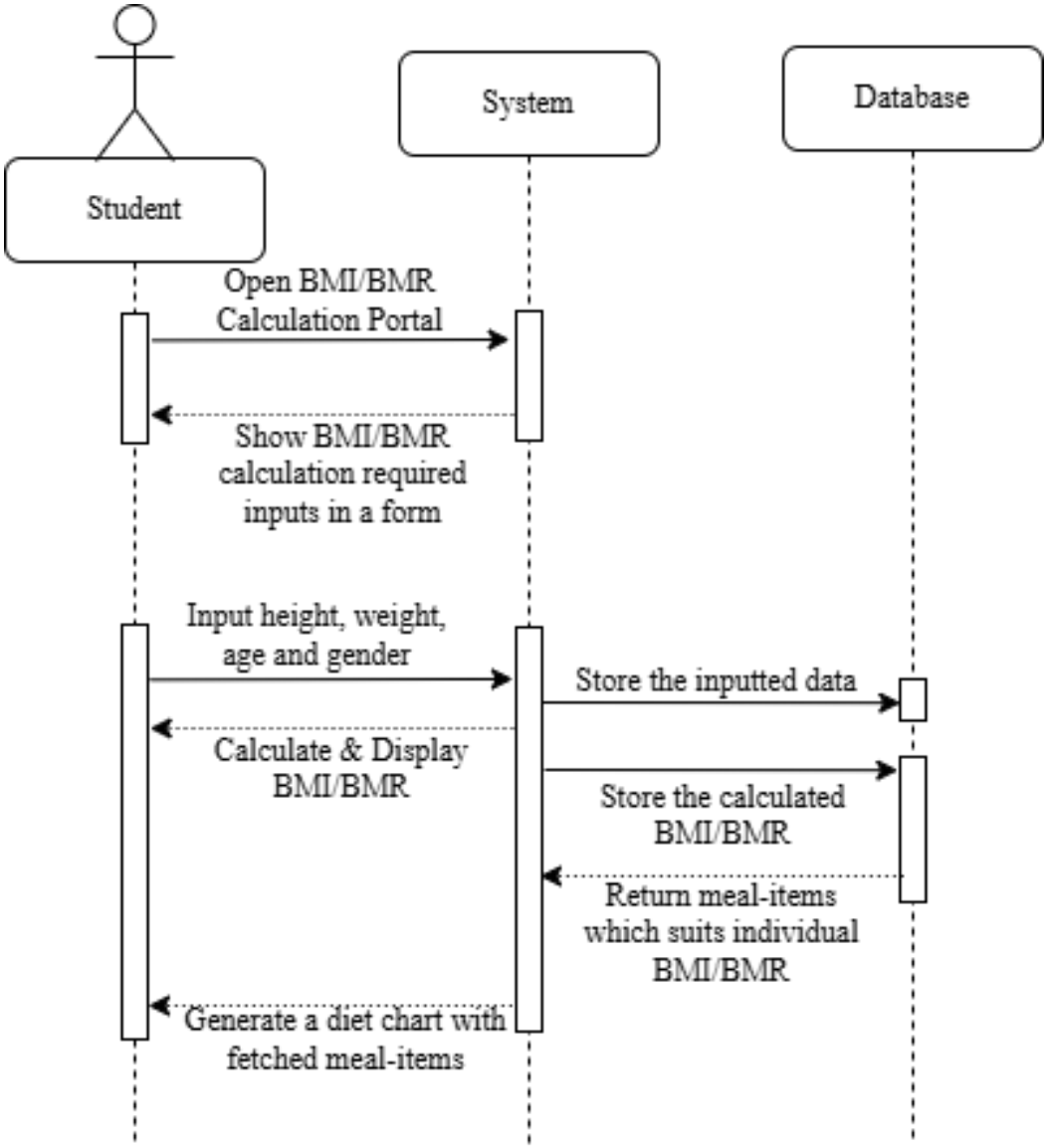
3.7.1 Student Register/Login:



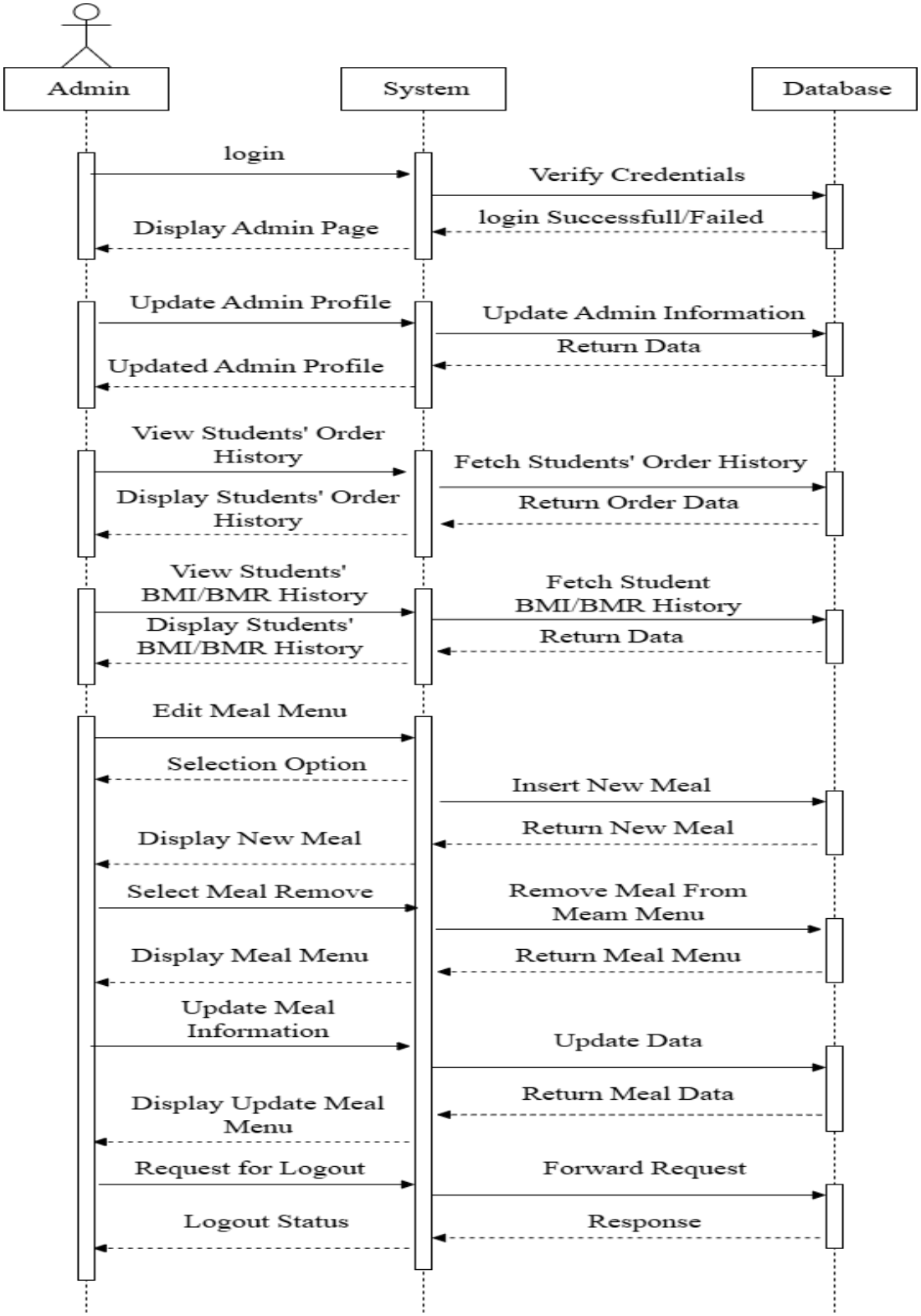
3.7.2 Meal-Menu Properties:



3.7.3 BMI/BMR Calculation properties:



3.7.4 Admins' Functionalities:



3.8 Conclusion:

The Students' Daily Meal Management System simplifies the process of managing students' daily meals and tracking students' health. With features like admin login, meal menu editing, order history, and BMI/BMR monitoring, it meets key functional needs while ensuring performance, security, and usability.

Designed for scalability and reliability, the system supports efficient decision-making and promotes students' wellness through a user-friendly interface and secure data handling. Overall, it offers an effective and modern solution for enhancing nutrition management in educational institutions.

Chapter 4

Architectural Design & Testing

4.1 Architectural Model:

An architectural model is a structured framework that defines the organization of a system's components, their relationships, and interactions to ensure scalability, maintainability, and clarity in software development. One widely adopted style is the ***Layered Architecture***, where the system is divided into horizontal layers, each with a specific role—from user interaction to data storage—ensuring clear separation of concerns. In the context of ***Nutri-Track Academia***, a students' daily meal management system, the layered architecture is an ideal fit due to the system's modular responsibilities: user interaction (students/admins), business logic (order processing, BMI/BMR computation), service coordination (authentication, session handling), and persistent data management (MySQL database). The system contains distinct user flows—students ordering meals or drinks and viewing personal health insights, and administrators editing menus or reviewing student activity—which benefit from being decoupled across well-defined layers. By adopting the layered model, Nutri-Track ensures each concern (UI, logic, database) remains isolated, which simplifies debugging, testing, and future scalability. PHP's compatibility with such modularization and XAMPP's full-stack simulation make this architecture both practical and future-proof for web deployment.



Figure 01: Architectural Diagram

4.2 Hardware and Software Tools:

Hardware:

For the server:

➢ CPU: Intel i5, Intel i3

➢ RAM: 8GB, 16GB

➢ Storage: 512 GB SSD

Software:

➢ Operating System: Windows 11

➢ Frontend: HTML, CSS, JavaScript

➢ Backend: PHP

➢ Visual Studio Code: For code development.

➢ XAMPP: For local server setup with Apache, MySQL.

➢ Browser: For testing and previewing the website.

➢ PhpMyAdmin: For managing the MySQL database.

➢ For diagram: Draw.io

4.3. Nutri-Track Academia – Component-Level Design

1. User Management Component

Handles student and admin registration, login, and profile updates. Implements role-based access control, distinguishing between students and cafeteria managers (admins).

2. Student Management Component

Allows students to sign up, receive an auto-generated meal card number and view their order history .

3. Admin Management Component

Enables admins to edit their profile, view all students' order histories (with search by meal card no.), and access the menu management system.

4. Menu Management Component

Allows admins to add, update, or delete menu items (breakfast, lunch, snacks, drinks). Ensures students see only currently available menu items when placing orders.

5. Ordering Component

Enables students to browse menus and place orders for meals (based on time: breakfast, lunch, snacks) and drinks. Each order is stored with timestamp and student meal card number.

6. BMI/BMR Component

Provides students with a calculator to input height, weight, age, and gender, and returns BMI/BMR values along with a recommended diet chart based on results.

7. Order History Component

Lets students view their past orders, while admins can search and filter orders from all students by date or meal card number.

8. Search Component

Enables admins to search order records using a student's meal card number, allowing quick lookup of meal and drink history for specific users.

9. Database Component

Handles secure storage and retrieval of all data, including:

- Student information (student data)
- Admin information (admin data)
- Menus (breakfast_menu, lunch_menu, etc.)
- order_info (all order transactions)

Connected via XAMPP MySQL backend in the nutri_track database.

10. Security Component

Ensures authentication, role-based authorization, and encryption of sensitive data, such as login credentials and student meal card numbers. Protects data during order transactions and profile updates.

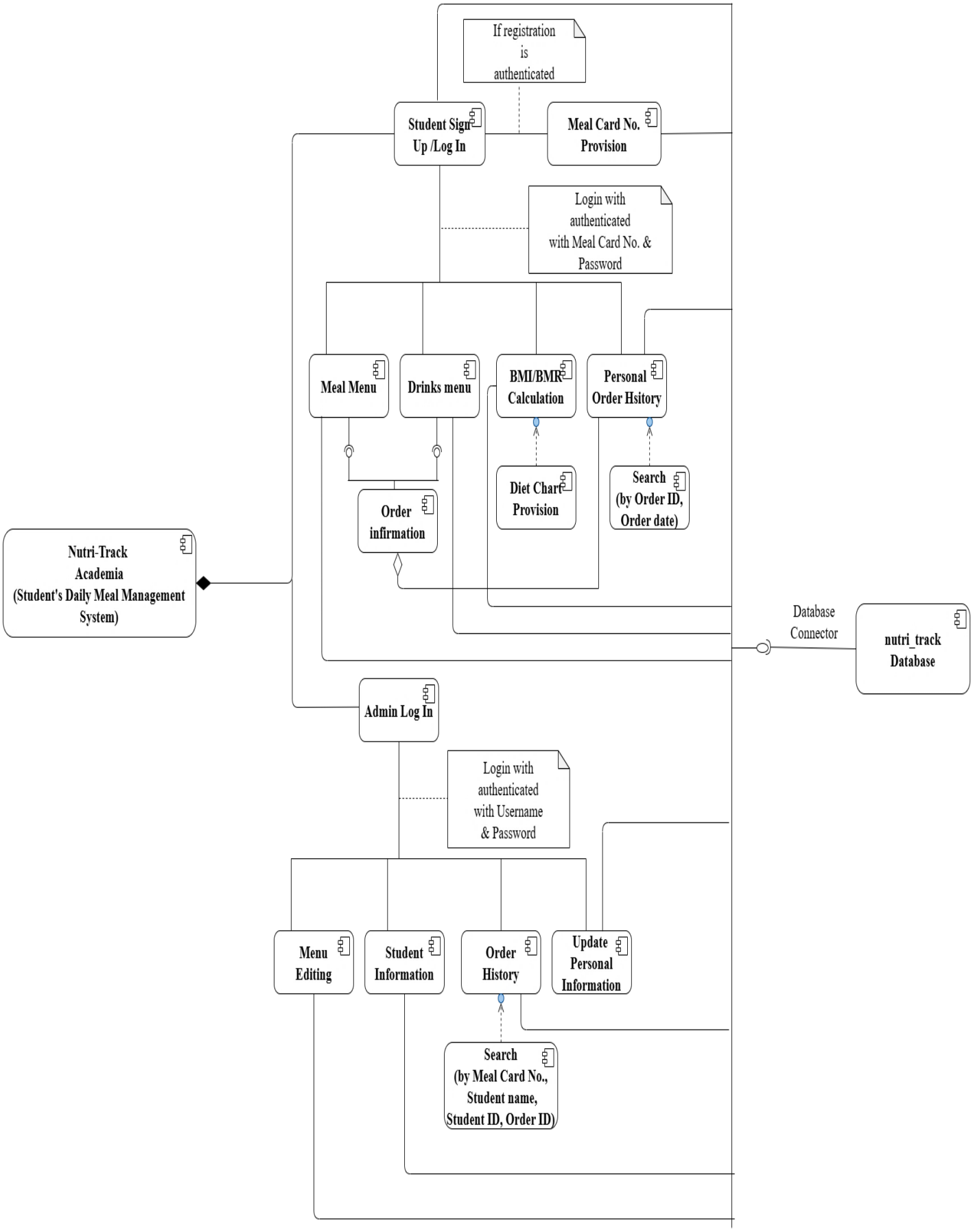


Figure 02: Structural Diagram

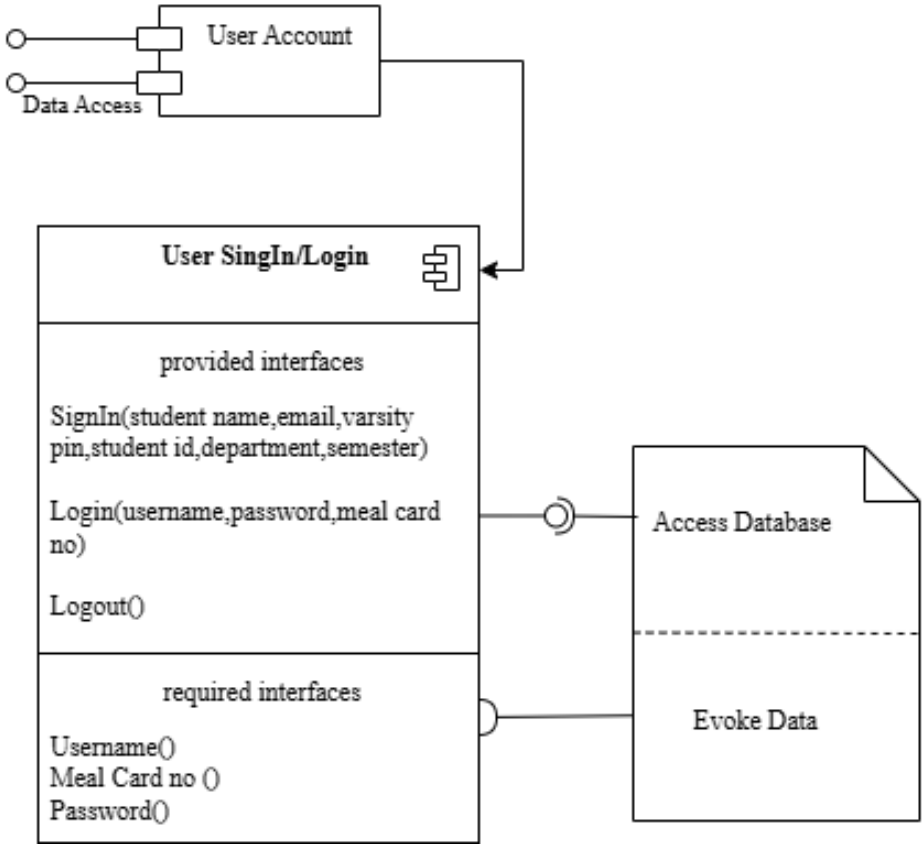


Figure 03: User Account Component

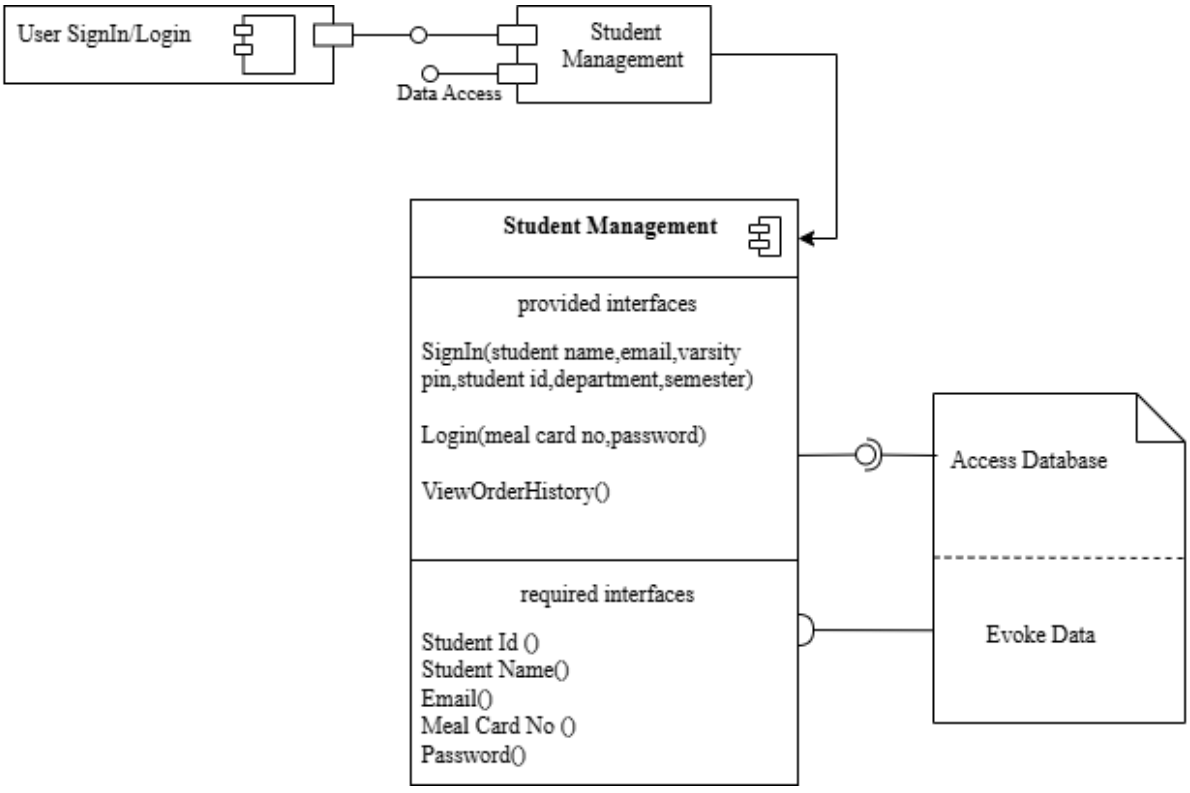


Figure 04: Student Management Component

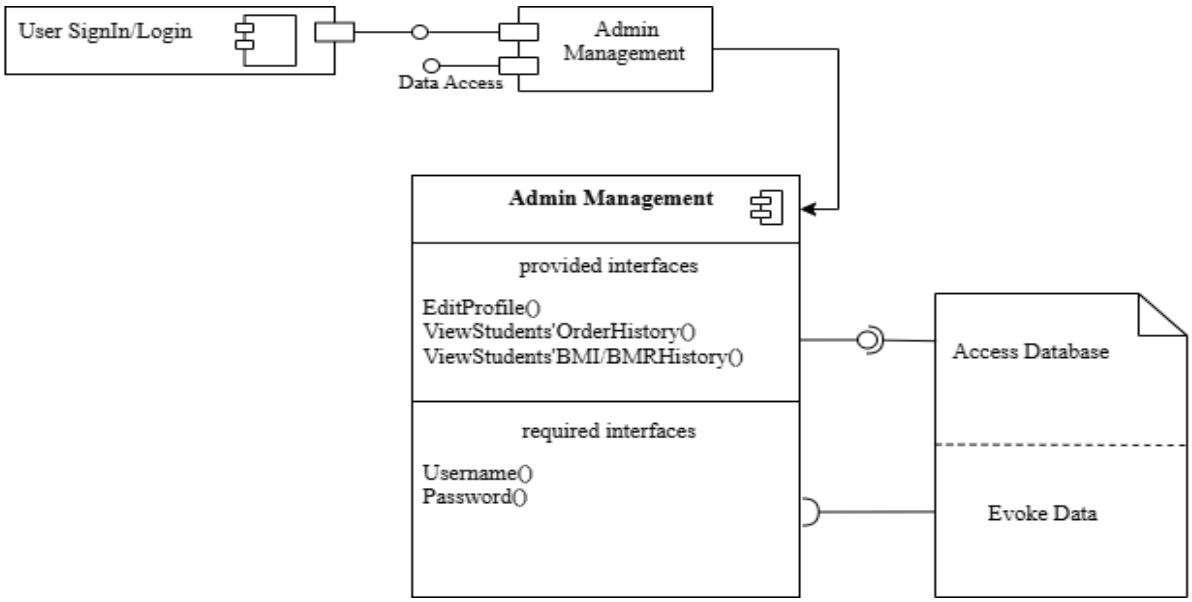


Figure 05: Admin Management Component

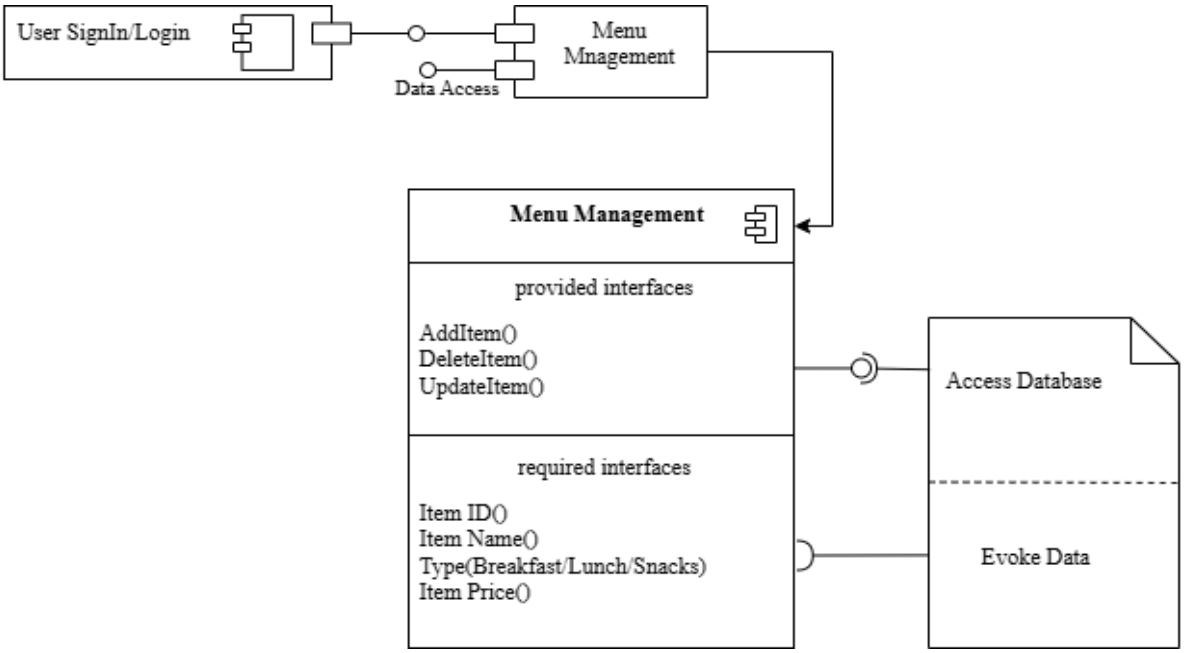


Figure 06: Menu Management Component

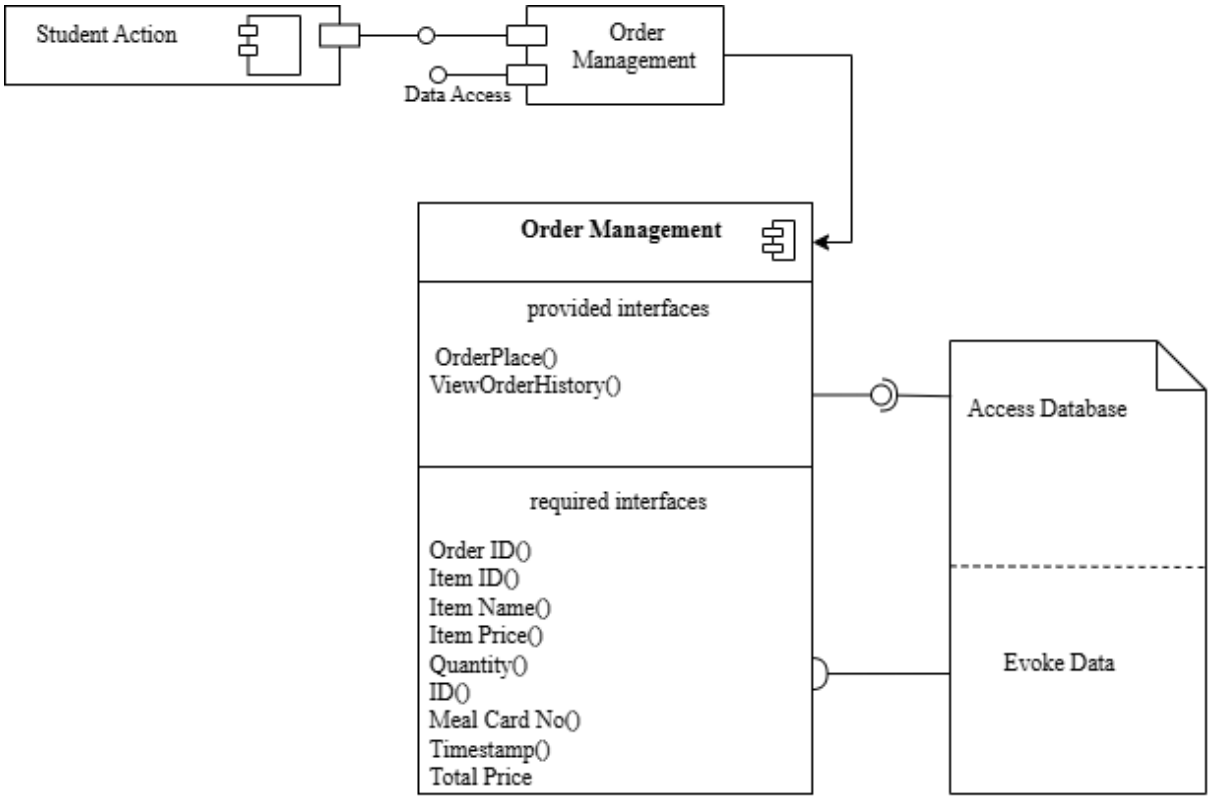


Figure 07: Order Management Component

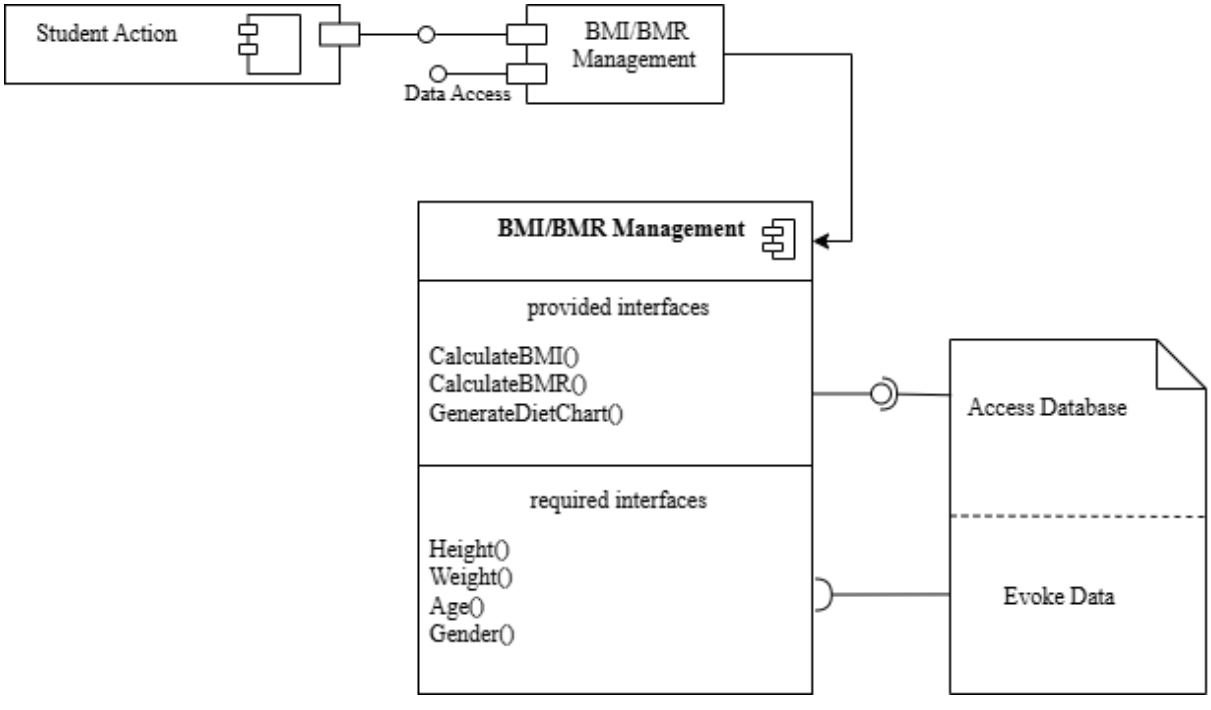


Figure 08: BMI/BMR Management Component

4.4. Graphical User Interface (GUI):

A Graphical User Interface (GUI) is the visual layer of a software system that enables users to interact with its functions through graphical elements such as buttons, menus, forms, and icons—without needing to understand the underlying code or logic. An effective GUI bridges the gap between human and machine, focusing on usability, accessibility, and efficiency. For ***Nutri-Track Academia***, the GUI plays a pivotal role in simplifying daily meal interactions for students and administrative staff. Designed with clean navigation and intuitive interfaces, students are presented with individual pages for logging in, ordering meals or drinks, calculating BMI/BMR for diet suggestions, and viewing their order history—all crafted to offer a seamless user experience. On the admin side, dedicated interfaces allow secure login, editing of the meal and drinks menu, viewing student order logs, and updating their personal profile. The GUI layout follows a consistent visual hierarchy, with responsive elements, input validations, and real-time feedback. Each page is tailored to its target user (student or admin), reducing cognitive load and guiding actions naturally. Built using HTML, CSS, and PHP, the GUI integrates tightly with backend logic while maintaining visual clarity and ease of use, supporting the goals of accessibility, efficiency, and system transparency.

The GUI of ***Nutri-Track Academia*** given below:

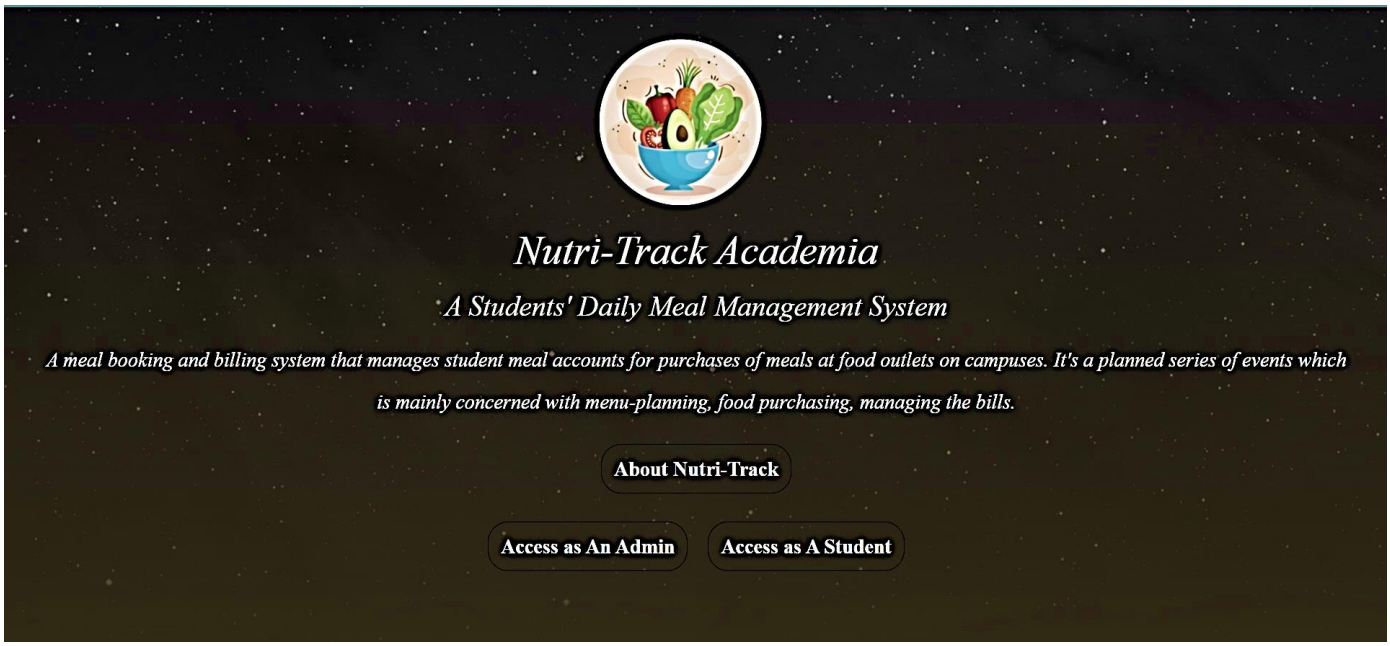


Figure 09: Home Page

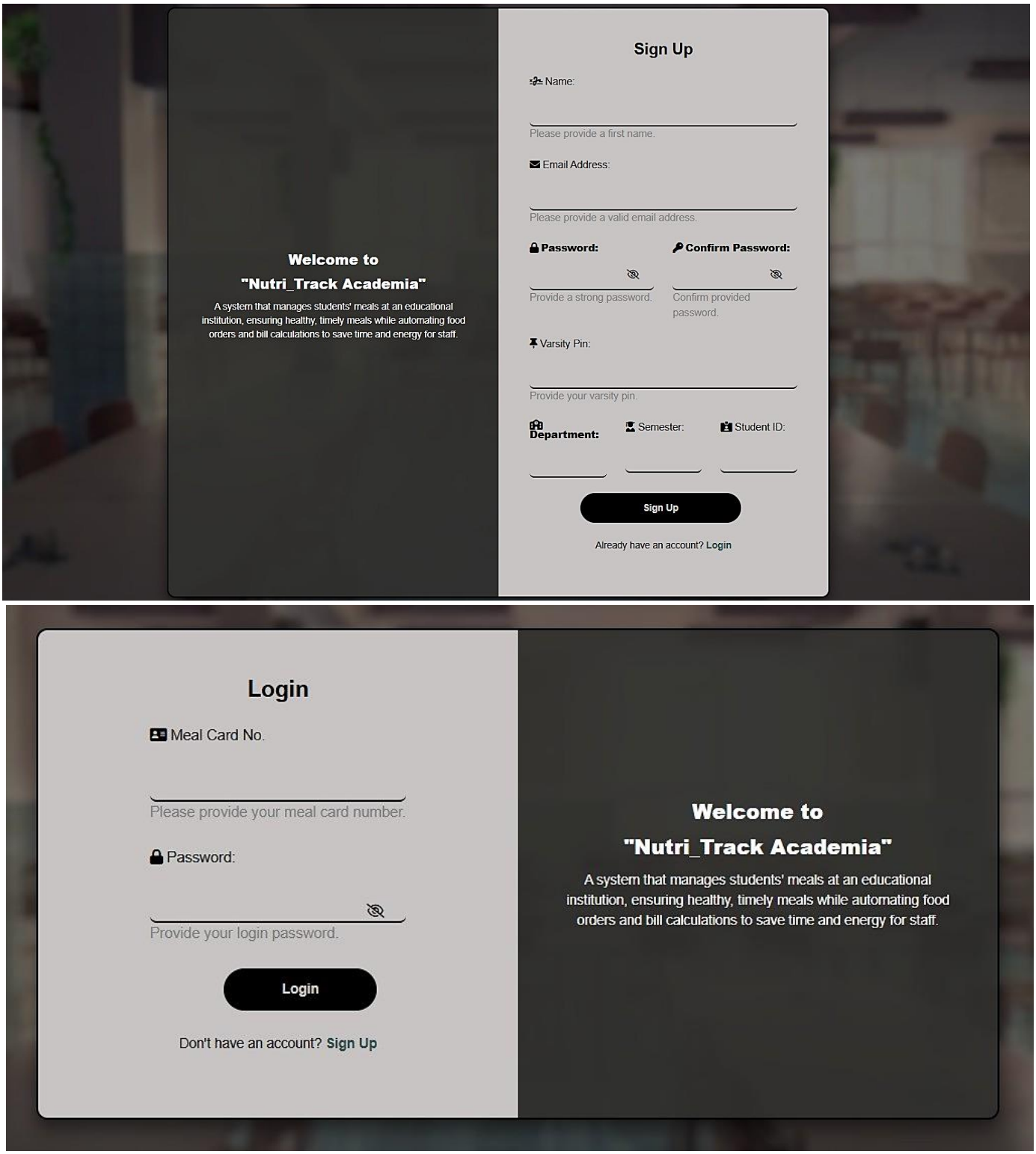


Figure 10: Student Sign Up/ Log In

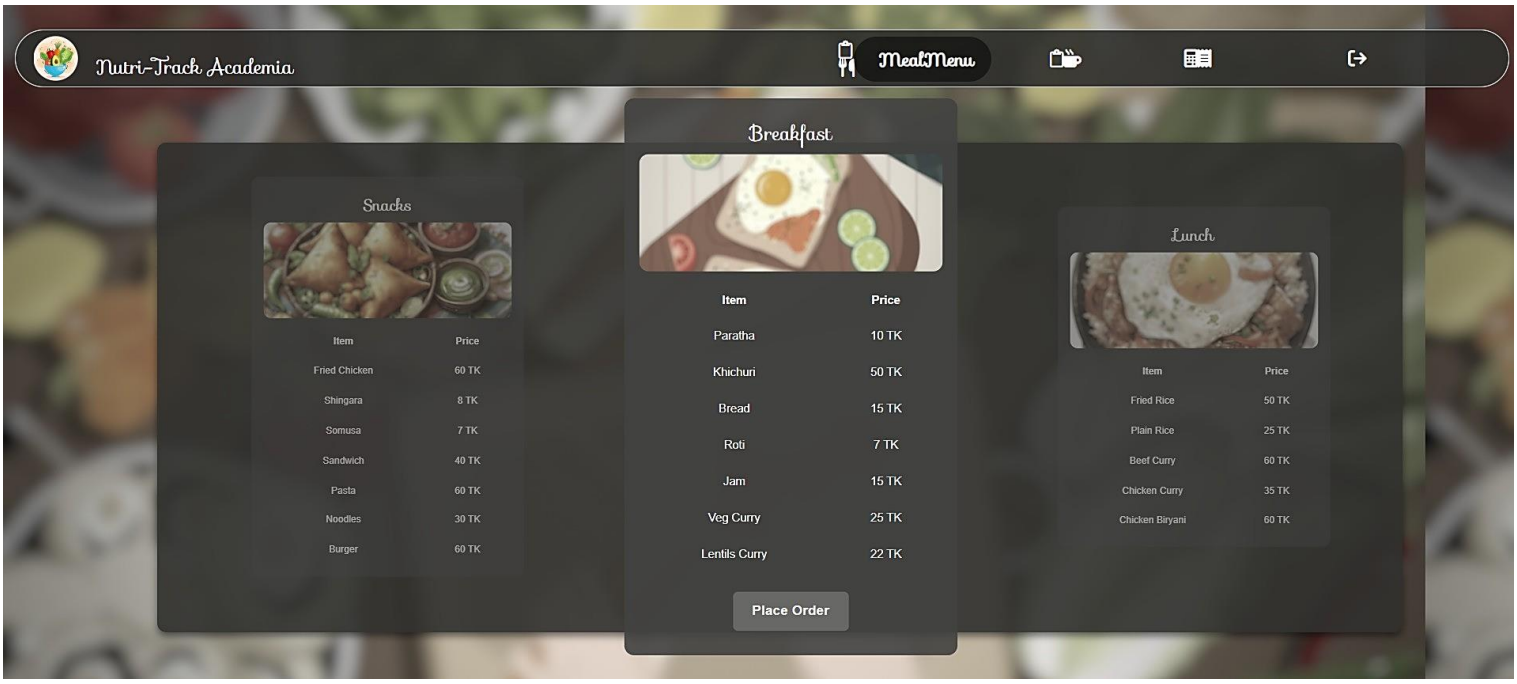


Figure 11: Student Viewing Meal menu based on time

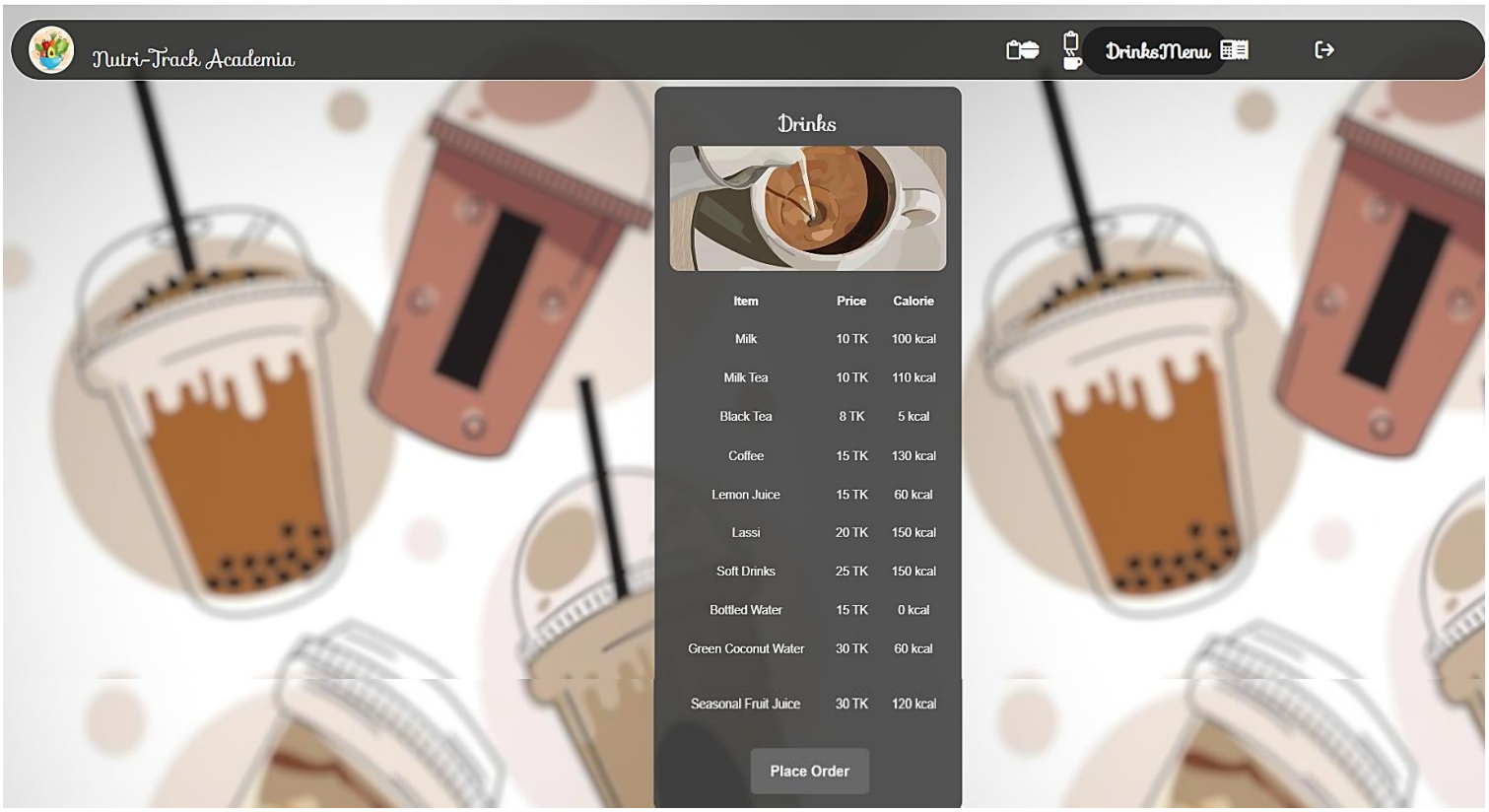


Figure 12: Student Viewing Drinks menu

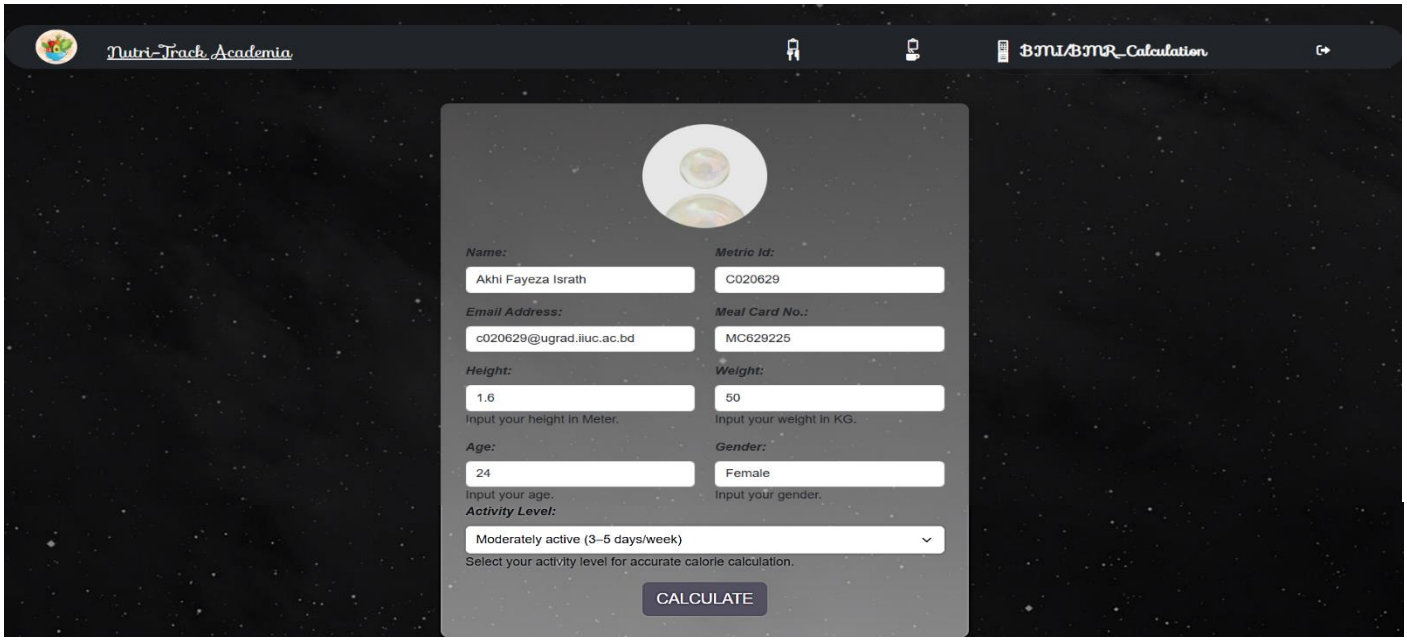
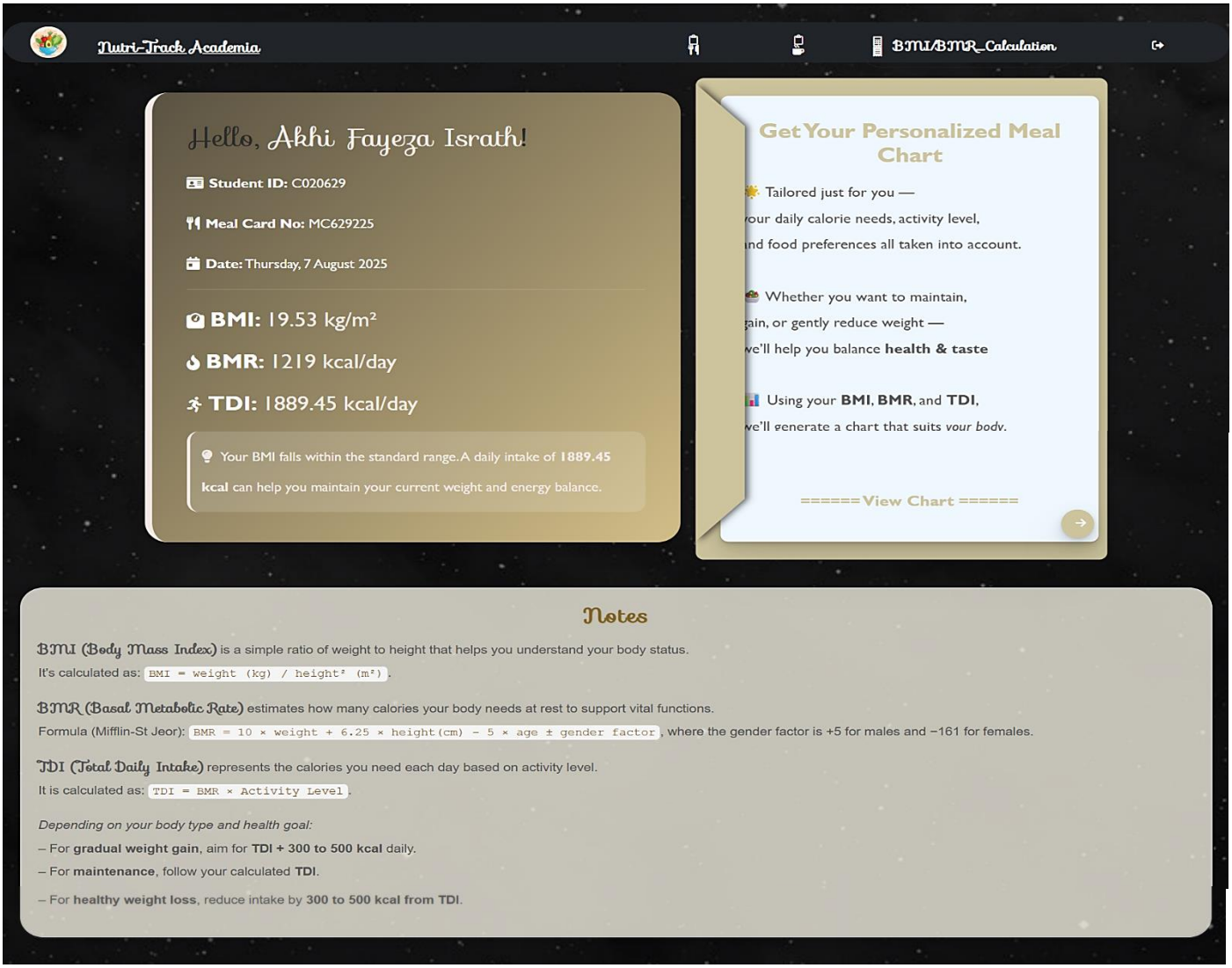


Figure 13: BMI/BMR Calculation page



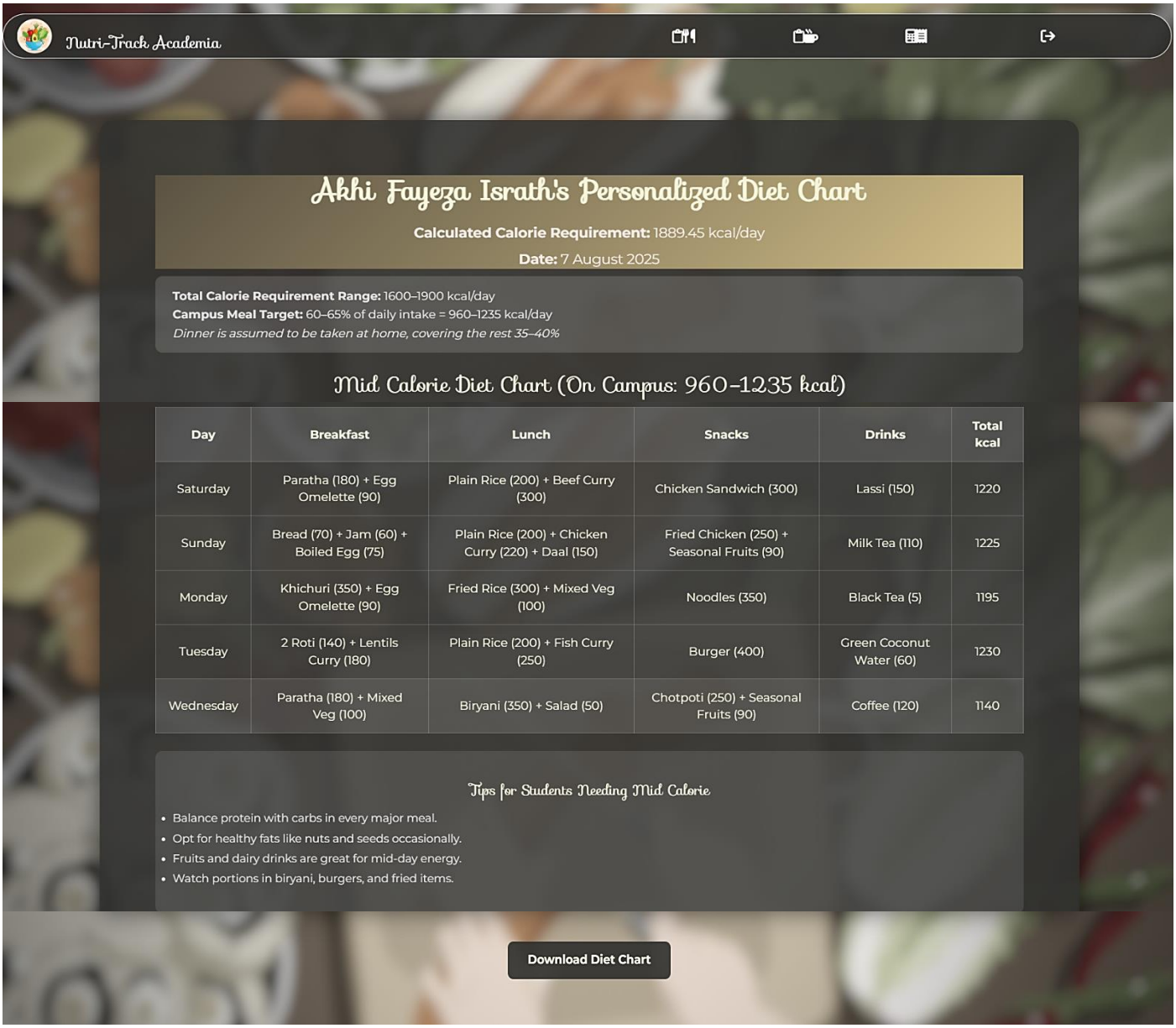


Figure 14: Diet Chart

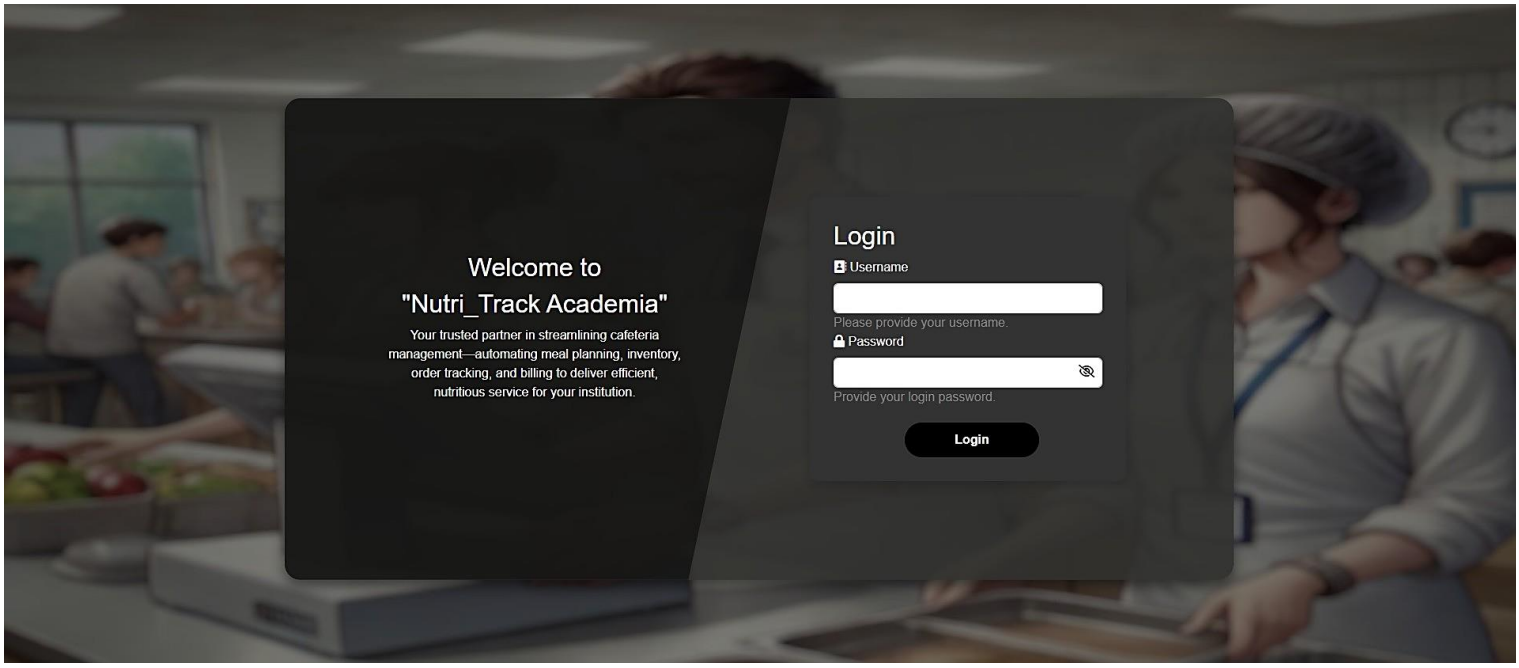


Figure 15: Admin Log In

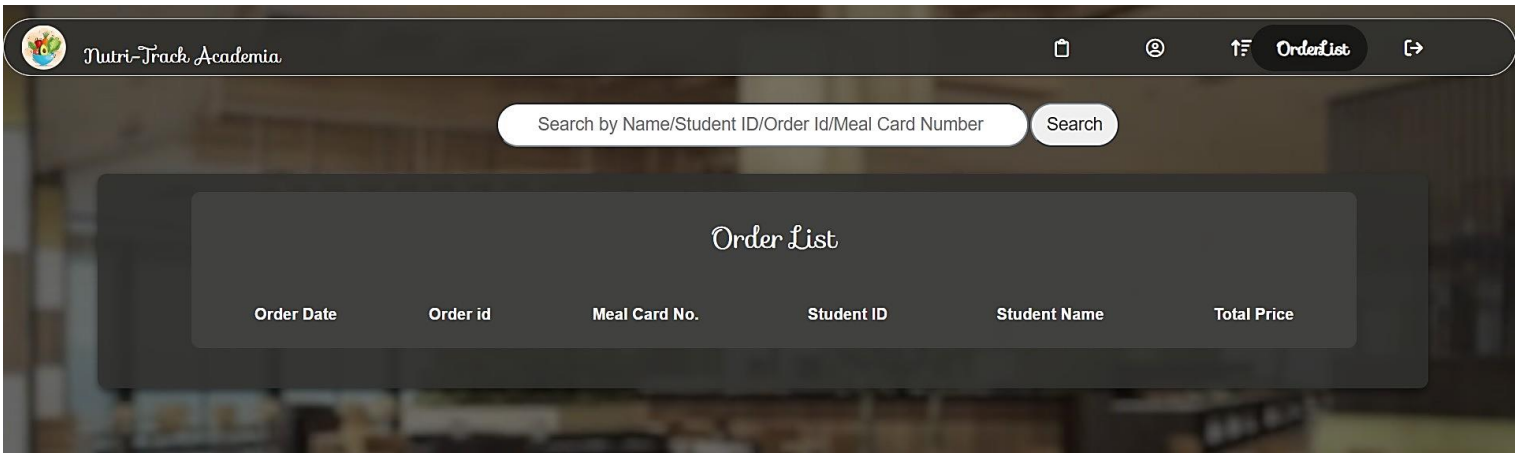


Figure 16: View Order History, Search

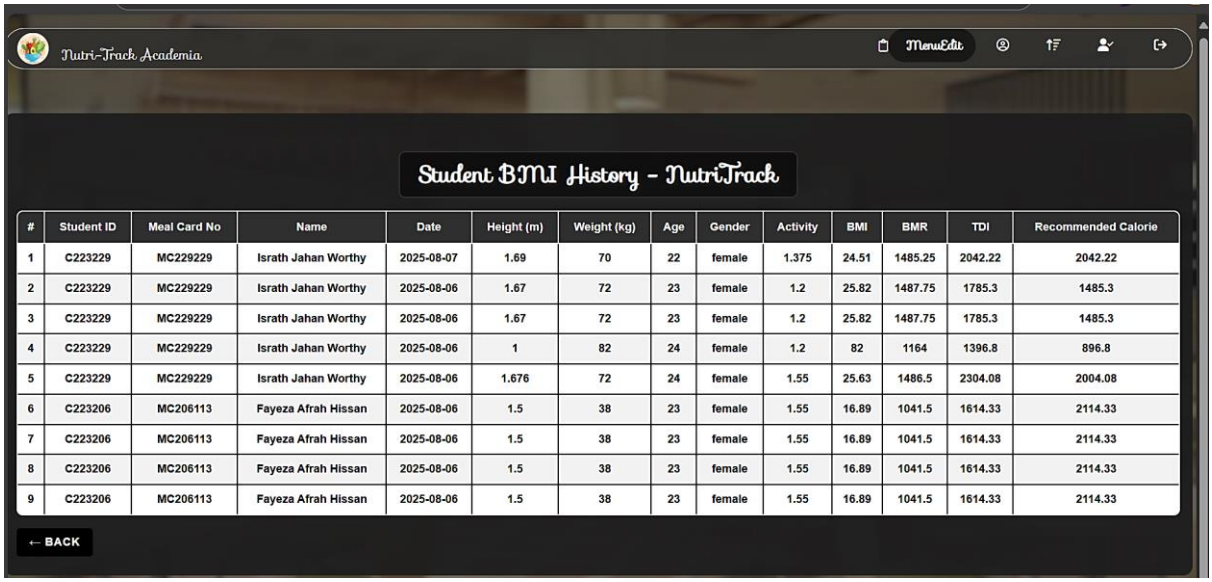


Figure 17: View Student BMI/BMR List

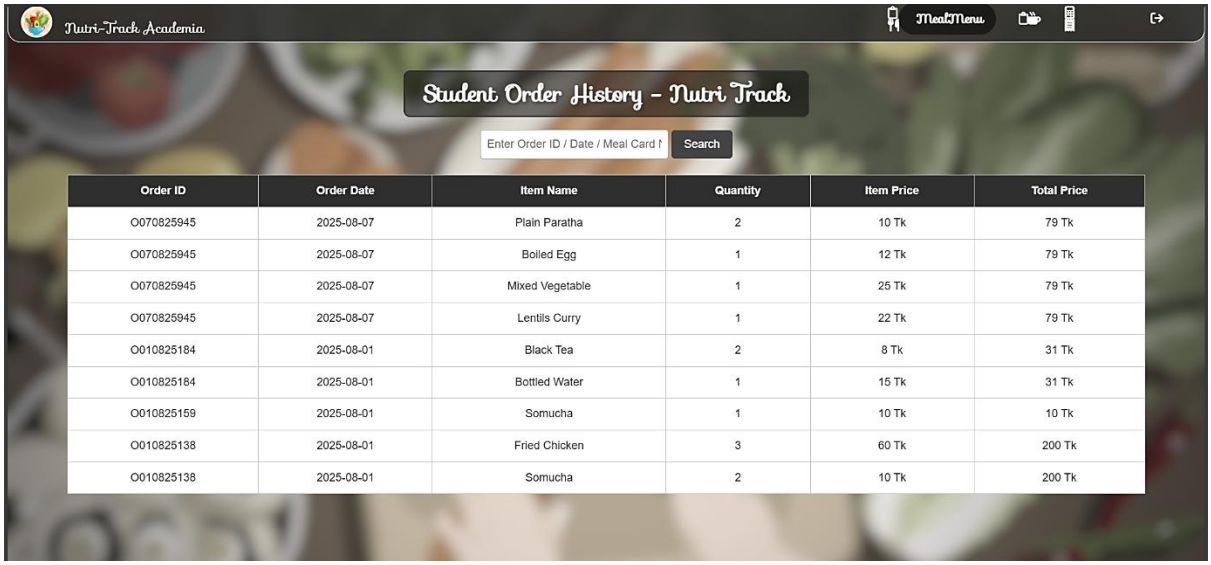


Figure 18: View personal order history, Search

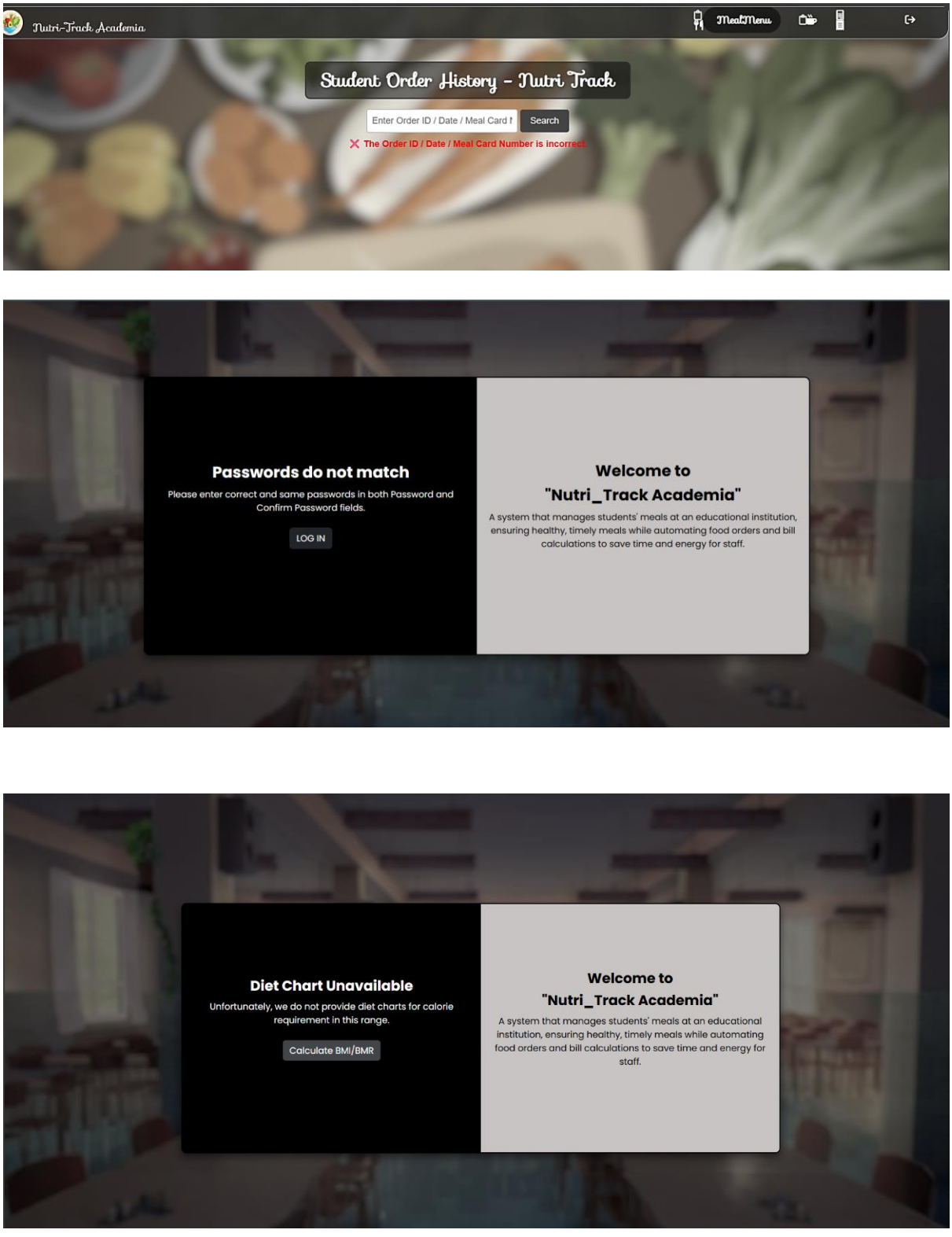


Figure 19: Some Error Messages

4.5. Requirement/Specifications-Based System Level Test Cases:

System level test cases are test cases based on the key requirements or specifications of a system. These test cases are derived directly from the overall functional and non-functional requirements of the system. These test the complete behavior of the system, ensuring it meets user expectations. Following table shows the requirements for our project, **Nutri-Track Academia – A Students' Daily Meal Management System** and the proper way to test if these requirements are fulfilled or not.

Table 1: Requirement/Specifications based system level test cases

Req. ID	Requirement Description	Test Case Description	Expected Outcome	Status
R1	User Registration with student details	Verify registration with valid data	User registered; meal card generated	Pass
R2	Login with meal card number and password	Verify login using valid credentials	Dashboard displayed	Pass
R3	View categorized menu (breakfast/lunch/snacks/ drinks)	Check menu is displayed per current time slot	Correct menu shown	Pass
R4	Ordering meals and bill calculation	Select meals and amounts to order	Correct bill is calculated based on given prices of meals and selected amounts	Pass
R5	See order details	Select meals and amounts to order and confirm order.	Order number, date and details are shown	Pass
R6	BMI/BMR calculation	Submit height, weight, age	BMI/BMR calculated with diet recommendations	Pass
R7	Admin login with username and password	Verify login using valid credentials	Dashboard displayed	Pass
R8	Admin profile update	Admin updates own profile info	Profile updated successfully	Pass
R9	Admin managing daily menus	Add, update, delete meals	Meal list updated accordingly	Pass
R10	Admin viewing order history and BMI/BMR list	Admin checking order history of all orders placed till date and students' BMI/BMR list	Order history and BMI/BMR list shown correctly	Pass

4.6. Module-Based Test Cases:

These test cases are created for individual modules or components of the system (e.g., Login, Menu, BMI/BMR calculator). They focus on isolated functionality and help identify issues early during development. The multiple tables that are shown below tests different modules of **Nutri-Track Academia – A Students' Daily Meal Management System** to check if the system requirements are properly fulfilled in different cases using actual examples.

A) Student Registration Module:

Project Name	Nutri-Track Academia – A Students' Daily Meal Management System
Module Name	Student Registration Module
Date of Creation	02/05/25
Date of Review	03/05/25

Table 2: Test Cases for Student Registration Module

Test Case ID	Scenario	Steps	Test Data	Expected Result	Actual Result	Status
TC-SR-01	Valid registration	Fill form → Submit	**Name:** Fayeza Afrah Hissan **Email Address:** c223206@ugrad.iiuc.ac.bd **Password:** Fayeza123 **Confirm Password:** Fayeza123 **Varsity Pin:** VP221113 **Department:** CSE **Semester:** 6 **Student ID:** C223206	Registered, meal card generated	Registered, meal card generated	Pass
TC-SR-02	Missing required fields	Leave fields empty → Submit	Blank inputs	Cannot submit	Cannot submit	Pass
TC-SR-03	Duplicate student ID	Register again with same ID	**Student ID:** C223206	Error: ID already exists	Error: ID already exists	Pass
TC-SR-04	Non existing student information	Register with student details which are not in the	**Name:** Tanisha Khan **Email Address:** c252123@ugrad.iiuc.ac.bd **Password:** Tanisha123	Error: Invalid Info Provided. (Information not	Error: Invalid Info Provided. (Information not	Pass

		varsity database	**Confirm Password:** Tanisha123 **Varsity Pin:** VP250101 **Department:** EEE **Semester:** 1 **Student ID:** E252123	found in student details database)	found in student details database)	
TC-SR-05	Wrong student information	Register with student details which do not match with the details in varsity database	**Name:** Israth Jahan Worthy **Email Address:** c223229@ugrad.iiuc.ac.bd **Password:** Israth@29 **Confirm Password:** Israth@29 **Varsity Pin:** VP220000 (wrong) **Department:** CSE **Semester:** 6 **Student ID:** C223229	Error: Invalid Info Provided. (Since varsity pin does not match with that stored in database)	Error: Invalid Info Provided. (Since varsity pin does not match with that stored in database)	Pass
TC-SR-06	Too short password length	Register with a password less than 8 characters	**Password:** Akhi	Cannot submit due to password length being shorter than 8 characters	Cannot submit due to password length being shorter than 8 characters	Pass
TC-SR-07	Password and Confirm Password fields do not match	Register with different values in Password and Confirm Password fields	**Password:** Akhi1234 **Confirm Password:** Akhi1111	Error: Passwords do not match.	Error: Passwords do not match.	Pass

B) Login Module:

Project Name	Nutri-Track Academia – A Students' Daily Meal Management System
Module Name	Student Registration Module
Date of Creation	04/05/25
Date of Review	10/05/25

Table 3: Test Cases for Login Module

Test Case ID	Scenario	Steps	Test Data	Expected Result	Actual Result	Status
TC-LG-01	Valid login	Input meal card no + password → Login	**Meal Card No:** MC229229, **Password:** israth@29	Login successful	Login successful	Pass
TC-LG-02	Invalid credentials	Input wrong meal card no → Login	**Meal Card No:** MC229000, **Password:** israth@29	Error: Invalid Meal Card No.	Error: Invalid Meal Card No.	Pass
TC-LG-03	Invalid credentials	Input wrong password → Login	**Meal Card No:** MC229229, **Password:** israth123	Error: Passwords do not match	Error: Passwords do not match	Pass

C) Menu Viewing & Ordering Module

Project Name	Nutri-Track Academia – A Students' Daily Meal Management System
Module Name	Student Registration Module
Date of Creation	10/05/25
Date of Review	19/05/25

Table 4: Test Cases for Menu Viewing & Ordering Module

Test Case ID	Scenario	Steps	Test Data	Expected Result	Actual Result	Status
TC-MN-01	View lunch menu at noon	Login → View Menu	**Time:** 2 PM	Lunch menu displayed	Lunch menu displayed	Pass
TC-MN-02	View no menu at wrong time	Login → View Menu	**Time:** 7 PM	No menu displayed	No menu displayed	Pass
TC-MN-03	Order item	Select item + quantity→ Confirm order	**Item:** Fried Rice **Quantity:** 1	Order confirmed	Order confirmed	Pass
TC-MN-04	Proper bill calculation	Select items + quantity	**Item:** Fried Rice (Price=TK.50)	Total Price: TK.195	Total Price: TK.195	Pass

			Quantity: 2 **Item:** Beef Curry (Price=TK.60) **Quantity:** 1 **Item:** Chicken Curry (Price=TK.35) **Quantity:** 1			
TC-MN-05	See order details	Select item + quantity→ Confirm order	**Item:** Fried Rice **Quantity:** 1	Show student name, meal card number, order number, order date and order details (item name, price, quantity, subtotal, total price)	Show student name, meal card number, order number, order date and order details (item name, price, quantity, subtotal, total price)	Pass

D) BMI/BMR Module

Project Name	Nutri-Track Academia – A Students' Daily Meal Management System
Module Name	Student Registration Module
Date of Creation	18/06/25
Date of Review	19/06/25

Table 5: Test Cases for BMI/BMR Module

Test Case ID	Scenario	Steps	Test Data	Expected Result	Actual Result	Status
TC-BM-01	Calculate BMI/BMR	Input age, gender, weight, height → Calculate	**Height:** 1.5 m **Weight:** 55 kg **Gender:** Female **Age:** 24	BMI/BMR result shown. BMI: 24.44 BMR: 1658.94 calories/day	BMI/BMR result shown. BMI: 24.44 BMR: 1658.94 calories/day	Pass
TC-BM-02	Invalid input	Input text instead of number	**Weight:** abc	Error: Invalid input	Error: Invalid input	Pass

			Height: xyz			
TC-BM-03	Get diet chart	Input age, gender, weight, height → Calculate	**Height:** 1.5 m **Weight:** 55 kg **Gender:** Female **Age:** 24	BMI/BMR result shown with personalized diet chart according to it	BMI/BMR result shown with personalized diet chart according to it	Pass

E) Admin Panel Module

Project Name	Nutri-Track Academia – A Students' Daily Meal Management System
Module Name	Student Registration Module
Date of Creation	24/05/25
Date of Review	30/05/25

Table 6: Test Cases for BMI/BMR Module

Test Case ID	Scenario	Steps	Test Data	Expected Result	Actual Result	Status
TC-AD-01	Valid login	Input username + password → Login	**Username:** faija@8 **Password:** ntac@194	Login successful	Login successful	Pass
TC-AD-02	Invalid credentials	Input username (wrong) + password → Login	**Username:** faija@1234 **Password:** ntac@194	Error: Invalid Username	Error: Invalid Username	Pass
TC-AD-03	Invalid credentials	Input username + password (wrong) → Login	**Username:** faija@1234 **Password:** ntac@1234	Error: Passwords do not match	Error: Passwords do not match	Pass
TC-AD-04	Update profile	Change email → Save	**New Email Address:** amj13@gmail.com	Profile updated	Profile updated	Pass
TC-AD-05	Add menu item	Add new menu item details → Save	**New Item Id:** B07 **New Item Name:** Khichuri **Price:** TK. 25	Menu updated	Menu updated	Pass
TC-AD-06	Add menu item, but with	Add new menu item	**New Item Id:** B07	Error: Item Id	Error: Item Id	Pass

	existing menu id	details → Save		already exists	already exists	
TC-AD-07	Add menu item but with invalid data type in price	Add new menu item details → Save	**Price:** abc	Cannot input anything other than numbers	Cannot input anything other than numbers	Pass
TC-AD-08	Delete menu item	Go to menu and select item → Delete	N/A	Menu item deleted	Menu item deleted	Pass
TC-AD-09	Update menu item	Go to menu and select item + edit details → Update	**Item Id:** B02 **Item Name:** Bread **Price:** 10 (new)	Menu item updated	Menu item updated	Pass
TC-AD-10	View all student orders	Go to order history	N/A	Orders listed	Orders listed	Pass
TC-AD-11	View all student BMI/BMR	Go to BMI/BMR list	N/A	BMI/BMR listed	BMI/BMR listed	Pass

4.7. Traceability of Test Cases to Use Cases:

Traceability of test cases to use cases refers to the ability to directly link specific test cases in the software testing process to their corresponding user scenarios (use cases). A traceability matrix is a table that maps each test case to the corresponding requirement ID. It ensures all requirements have been tested and helps track testing coverage and gaps. We create the following traceability matrices for test cases of different modules from the module-based testing to the specific requirement it fulfills, taken from the requirement/specifications-based system level testing.

Table 7: Traceability Matrix for Student Registration Module

Test Case ID	**Requirement ID**
	R1
TC-SR-01	✓
TC-SR-02	✓

TC-SR-03	✓
TC-SR-04	✓
TC-SR-05	✓
TC-SR-06	✓
TC-SR-07	✓

Table 8: Traceability Matrix for Login Module

Test Case ID	**Requirement ID**
	R2
TC-LG-01	✓
TC-LG-02	✓
TC-LG-03	✓

Table 9: Traceability Matrix for Menu Viewing & Ordering Module

Test Case ID	**Requirement ID**		
	R3	R4	R5
TC-MN-01	✓		
TC-MN-02	✓		
TC-MN-03		✓	
TC-MN-04		✓	
TC-MN-05			✓

Table 10: Traceability Matrix for BMI/BMR Module

Test Case ID	**Requirement ID**
	R6
TC-BM-01	✓
TC-BM-02	✓
TC-BM-03	✓

Table 11: Traceability Matrix for Admin Panel Module

Test Case ID	**Requirement ID**			
	R7	R8	R9	R10
TC-AD-01	✓			
TC-AD-02	✓			
TC-AD-03	✓			
TC-AD-04		✓		
TC-AD-05			✓	
TC-AD-06			✓	
TC-AD-07			✓	
TC-AD-08			✓	
TC-AD-09			✓	
TC-AD-10				✓
TC-AD-11				✓

4.8. Techniques Used for Test Generation:

For the **Nutri Track Academia – A Students' Daily Meal Management System**, multiple test generation techniques were used to ensure thorough validation of system functionalities and to uncover both expected and edge-case behaviors. These techniques include:

1. Use Case-Based Testing

- **Purpose**: To ensure that all major system functions (like Registration, Login, Viewing Menus, Placing Orders, etc.) are correctly implemented.
- **Example**: Test cases such as TC-SR-01 (Student Registration Module) and TC-MN-04 (Placing meal orders) were derived directly from the use cases of the system.

2. Positive and Negative Testing

- **Purpose**: To check how the system handles both valid and invalid inputs.
- **Example**: Login test cases checked for correct credentials (positive) and incorrect or blank inputs (negative), ensuring robust validation mechanisms.

3. Equivalence Partitioning (EP)

- **Purpose**: To reduce the number of test cases by dividing input data into valid and invalid partitions.
- **Example**: Meal card numbers, emails, and numeric fields (like weight/height in BMI calculator) were tested using representative values from each valid/invalid partition.

4. Boundary Value Analysis (BVA)

- **Purpose**: To test edge values at the limits of input fields.
- **Example**: Tested BMI/BMR inputs with minimum and maximum allowed values for weight and height to ensure proper error handling or acceptance.

5. State Transition Testing

- **Purpose**: To verify system behavior during state changes.
- **Example**: Logging in as a student and transitioning to menu browsing or order placing; or logging out and returning to the unauthenticated state.

6. Error Guessing

- **Purpose**: Based on experience, guessing potential user errors or system misbehaviors.
- **Example**: Predicting user attempts to place orders without logging in or entering letters in number fields during registration.

7. Exploratory Testing

- **Purpose**: Manual unscripted testing to uncover hidden issues and usability flaws.
- **Example**: Navigating through the student interface as a new user to find misalignments, missing validations, or unexpected behavior in modules like BMI calculation or viewing history.

Chapter 5

Conclusion

5.1 Limitations of Work

• **Basic Authentication**: Currently, the system supports only basic login functionalities without advanced security features like OTP, 2FA, or email verification.

• **Static Diet Chart**: Diet suggestions are generated using standard BMI/BMR calculations without considering individual health conditions or allergies.

• **Limited Order Customization**: Students cannot customize meals (e.g., desired meals, ingredient preference) — only fixed menu items can be ordered.

• **No Payment Integration**: The system calculates total prices but lacks real payment gateway integration for online transactions.

• **No Notification System**: The system lacks email/SMS or in-app notifications for order confirmations, diet chart updates, or admin announcements.

• **Limited Analytics**: Admins can view data but lack advanced analytics like most-ordered meals, student health trends, or nutritional gaps.

5.2 Future Work

• **Mobile Application**: Develop mobile versions (iOS/Android) for easier student access and real-time ordering.

• **Secure Online Payments**: Add online payment options through platforms like Nagad, Rocket, or bKash for prepaid meal ordering.

• **Dietitian Dashboard**: Add a panel for certified dietitians to give personalized suggestions, track student health metrics, and approve diets.

• **Student Feedback System**: Allow students to rate meals, suggest improvements, and report issues.

• **Meal Subscription Plans**: Introduce weekly/monthly subscription packages with discounted prices.

• **Integration with Smart Cards/QR Codes**: Use QR codes or smart ID cards for meal pickup authentication.

• **Admin Analytics Dashboard**: Implement charts and reports to monitor food trends, student health stats, and system performance.